# STA302 — Methods of Data Analysis

Course Notes

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# Week 1: Course Overview and Simple Linear Regression

### Learning Objectives

*Basic* Distinguish population vs. sample quantities, random variables vs. fixed values §; calculate LS coefficient estimates §2; interpret estimated coefficients in context §1.

*Adv.* Explain the LS process to a broad audience; derive LS estimates for various model specifications; apply SLR in R (`lm()`).

## Regression Basics

**Definition - Regression**

$Y \approx f(X_1, \ldots, X_p)$, $p \geq 1$ predictors. $Y$ = response; $X_i$ = predictors/covariates/features.

- **Simple** ($p = 1$) vs **Multiple** ($p > 1$).
- **Parametric**: $f$ has a fixed parametric form, e.g. $\beta_0 + \beta_1 X$, $\beta_0 + \beta_1 X + \beta_2 X^2$. Else **non-parametric** (kNN, trees, neural nets).

## The SLR Model

**Definition - SLR Model**

$$y_i = \beta_0 + \beta_1 x_i + \varepsilon_i, \quad i = 1, \ldots, n, \quad E[\varepsilon_i] = 0 \ \forall i$$

- $\beta_0, \beta_1$: fixed, unknown parameters — *same* for every $i$.
- $\varepsilon_i$ ($= e_i = \epsilon_i$): random, zero-mean — *differs* by $i$.

*Was:* $y_i \overset{iid}{\sim} N(\mu, \sigma^2)$. *Now:* $y_i \mid x_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

**Functional** ($y_i = \beta_0 + \beta_1 x_i$, no error) vs **statistical** relationship ($y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$); $\varepsilon_i = y_i - (\beta_0 + \beta_1 x_i)$ = residual.

**"Linear"** = linear in $\beta_0, \beta_1$, not in $X$.

- OK (still linear): $\beta_0 + \beta_1 x_i^2$, $\beta_0 + \beta_1 \sin x_i$, $\beta_0 + \beta_1 \mathbb{I}(\cdot)$ — free to transform $X$.
- NOT linear: $\log(\beta_0 + \beta_1 x_i + \varepsilon_i)$, $\beta_0 e^{\beta_1 x_i} + \varepsilon_i$ — *don't transform $Y$.*

**Always condition on $X$:** $E[Y \mid X] = \beta_0 + \beta_1 X$ (abbreviated $E[Y]$, $\mathrm{Var}[Y]$). *$E[Y]$ alone is meaningless — need e.g. $E[Y \mid X = 5]$.*

**Notation:** sample $(x_i, y_i)$ = realizations of $(X, Y)$; population $Y = \beta_0 + \beta_1 X + \varepsilon$.

## Fitted Model & Predictions

**Definition - Fitted SLR**

$\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, where $\hat{\beta}_0 \to \beta_0$, $\hat{\beta}_1 \to \beta_1$, $\hat{Y} \to E[Y \mid X]$.

**Example 1 Credit Score**

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Income ($X$, \$100k) | 1.3 | 1.1 | 0.8 | 1.2 | 1.4 | 0.9 | 0.9 | 1.4 | 1.2 | 1.0 |
| Credit Score ($Y$) | 756 | 728 | 635 | 599 | 760 | 722 | 743 | 726 | 694 | 726 |

`lm(creditScore ~ income)`: $\hat{\beta}_0 = 632.38$, $\hat{\beta}_1 = 68.32$

$$\hat{Y} = 632.38 + 68.32X$$

**Interpretation:**

- $\hat{\beta}_0$ = avg. $Y$ when $X = 0$ *avg. credit score, \$0 income*
- $\hat{\beta}_1$ = avg. change in $Y$ per unit increase in $X$ *per \$100k income*
- Prediction: $\hat{Y} = \hat{\beta}_0 + \hat{\beta}_1 X$, e.g. $X = 1.05 \Rightarrow \hat{Y} = 704.1$

**Validity of predictions** — two independent checks:

- **Statistical**: is $X$ inside the training range? (else **extrapolation** — training range was $X \in [0.8, 1.4]$)
- **Subject-matter**: do $X, \hat{Y}$ make sense? (credit score $\notin [300, 900]$, or negative income $\Rightarrow$ invalid)

## Least Squares Estimation

$$RSS = \sum_{i=1}^{n} \hat{\varepsilon}_i^2 = \sum_{i=1}^{n} \left(y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i)\right)^2 \quad \text{— minimize over } \hat{\beta}_0, \hat{\beta}_1$$

*Why square: no +/− cancellation, differentiable, penalizes outliers more; geometrically $\hat{Y}$ = orthogonal projection of $Y$ onto model space.*

**Identities:**

$$\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y}) = \sum_{i=1}^{n} x_i y_i - n\bar{x}\bar{y}, \qquad \sum_{i=1}^{n}(x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - n\bar{x}^2$$

**Proposition 2 LS Estimators.**

$$\hat{\beta}_1 = \frac{SP_{xy}}{SS_{xx}} = \frac{\sum x_i y_i - n\bar{x}\bar{y}}{\sum x_i^2 - n\bar{x}^2} = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sum (x_i - \bar{x})^2} = \frac{\mathrm{Cov}(X, Y)}{\mathrm{Var}(X)}, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

*$SS_{xx}$ = "sum of squares for $x$"; $SP_{xy}$ = "sum of products between $x, y$".* No relationship $\Rightarrow \hat{\beta}_1 = 0$, $\hat{\beta}_0 = \bar{y}$.

*Derivation.* $L(\beta_0, \beta_1) = \sum (y_i - \beta_0 - \beta_1 x_i)^2$.

$$\frac{\partial L}{\partial \beta_0} = 0 \;\Rightarrow\; \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\frac{\partial L}{\partial \beta_1} = 0 \;\Rightarrow\; \sum x_i y_i = \hat{\beta}_0 \sum x_i + \hat{\beta}_1 \sum x_i^2$$

Sub. $\hat{\beta}_0$: $\sum x_i y_i - n\bar{x}\bar{y} = \hat{\beta}_1(\sum x_i^2 - n\bar{x}^2) \Rightarrow \hat{\beta}_1$ as above. □

*Formulas only valid for 1 untransformed predictor.*

**Example 3 Deriving for a Different Specification: No Intercept**
$y_i = \beta_1 x_i + \varepsilon_i \Rightarrow L(\beta_1) = \sum(y_i - \beta_1 x_i)^2$. $\frac{\partial L}{\partial \beta_1} = -2\sum x_i(y_i - \beta_1 x_i) = 0 \Rightarrow \hat{\beta}_1 = \frac{\sum x_i y_i}{\sum x_i^2}$. *Same method every time: write $L$, differentiate, set $= 0$, solve.*

**Proposition 4 Properties, if $\beta_0$ in model.**

$$\sum_{i=1}^{n} \hat{\varepsilon}_i = 0, \qquad \sum_{i=1}^{n} x_i \hat{\varepsilon}_i = 0$$

*2nd property: $X$ and residuals uncorrelated (regardless of $\beta_0$). Good fit = no pattern in residuals vs. $X$.*

*Proof.* Sub $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$ into both (and $\hat{\beta}_1 = SP_{xy}/SS_{xx}$ into the 2nd). □

# Week 2: Multiple Linear Regression (MLR) Models

| **Learning Objectives** | |
|---|---|
| *Basic* | Write an MLR model in matrix form §; compute LS estimates via $(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ §8; interpret coefficients, indicators, and interactions §. |
| *Adv.* | Derive $\hat{\boldsymbol{\beta}}$ via matrix calculus; encode categorical predictors with dummy variables; recognize interaction effects from non-parallel slopes. |

## Matrices (Review)

### Definition - Matrix

A rectangular array of elements in $n$ rows and $p$ columns (*dimension/size* $n \times p$): $n$ = number of rows (1 per *observation*), $p$ = number of columns (1 per *explanatory variable*). *This is exactly* tidy data *format: 1 row per observation, 1 column per variable.*

$$\mathbf{A} = (a_{ij})_{i=1,\ldots,n;\, j=1,\ldots,p} = \begin{pmatrix} a_{11} & \cdots & a_{1p} \\ \vdots & \ddots & \vdots \\ a_{n1} & \cdots & a_{np} \end{pmatrix}$$

**Square**: $n = p$. **Column vector**: $p = 1$. **Row vector**: $n = 1$.

### Definition - Transpose, Symmetric

If $\mathbf{A}$ is $n \times p$, its **transpose** $\mathbf{A}'$ (or $\mathbf{A}^T$) is $p \times n$ (rows $\leftrightarrow$ columns). $\mathbf{A}$ is **symmetric** if $\mathbf{A} = \mathbf{A}'$ (only possible if square).

### Example 5 Design Matrix and X′X

$$\mathbf{Y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix} \textit{ vector of responses}, \qquad \mathbf{X} = \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} \textit{ always a column of 1's}$$

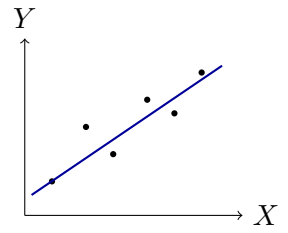
$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} 1 & 1 & 1 \\ x_1 & x_2 & x_3 \end{pmatrix} \begin{pmatrix} 1 & x_1 \\ 1 & x_2 \\ 1 & x_3 \end{pmatrix} = \begin{pmatrix} 3 & \sum x_i \\ \sum x_i & \sum x_i^2 \end{pmatrix}$$

$\mathbf{X}'\mathbf{X}$ is always square and symmetric.

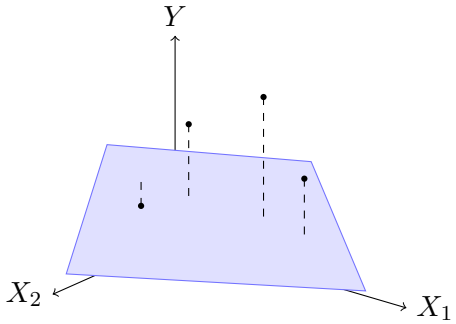
## The MLR Model

### Definition - Simple vs. Multiple Regression

$p = 1$: **Simple** Regression. $p > 1$: **Multiple** Regression, $f : \mathbb{R}^p \to \mathbb{R}$ — the more general form (SLR is MLR with $p = 1$).




*1 predictor: line of best fit*




*2 predictors: plane in* $\mathbb{R}^3$

*$p > 2$ predictors: hyperplane in $\mathbb{R}^{p+1}$ (can't visualize directly)*

**Definition - MLR Model**

For individual $i = 1, \ldots, n$:

$$Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i, \qquad \epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$$

where $\mathbf{x}_i' = (1, x_{i1}, \ldots, x_{ip})$ is treated as fixed/known. Stacking over $i$: $\boxed{\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}}$,

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1' \\ \vdots \\ \mathbf{x}_n' \end{pmatrix} = \begin{pmatrix} 1 & x_{11} & \cdots & x_{1p} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & x_{n1} & \cdots & x_{np} \end{pmatrix}, \quad \boldsymbol{\beta} = \begin{pmatrix} \beta_0 \\ \vdots \\ \beta_p \end{pmatrix}, \quad \boldsymbol{\epsilon} = \begin{pmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{pmatrix} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$$

*$\sigma^2\mathbf{I}_n$: multivariate normal, diagonal covariance* $\mathrm{diag}(\sigma^2, \ldots, \sigma^2)$ *— same $\sigma^2$, i.i.d. as in SLR.* $E[Y \mid \mathbf{X} = \mathbf{x}] = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$, $\mathrm{Var}[Y \mid \mathbf{X} = \mathbf{x}] = \sigma^2$ (again abbreviated $E[Y], \mathrm{Var}[Y]$, always conditioning on $\mathbf{X}$).

*$\mathbf{x}$ is $(p+1)$-dimensional: 1 intercept + $p$ predictors. (Some sources instead say $p$ entries total, $p-1$ predictors — check convention.)*

*Remark* (Design Matrix Conventions).
- First column is **always** 1's (intercept).
- Must be numeric — categorical predictors need **indicator (dummy) variables**, a.k.a. *one-hot encoding*.
- Each observation must fall in exactly 1 category.

**Example 6 Encoding a Categorical Predictor**

**2 levels** (e.g. transmission, automatic/manual): 1 indicator suffices, $X = \mathbb{I}(\text{manual})$.

$c$ **levels** (e.g. colour Blue/Red/Yellow): pick 1 **baseline** (in R: first level alphanumerically), need $c-1$ indicators. $X_2 = \mathbb{I}(\text{Red})$, $X_3 = \mathbb{I}(\text{Yellow})$; both 0 $\Rightarrow$ Blue.

**Example 7 Dimension of X**

7 numeric predictors, $n = 45$: $\mathbf{X}$ is $45 \times 8$ (1 intercept + 7 predictors).

1 continuous predictor + 1 categorical predictor with 10 levels, $n = 100$: $\mathbf{X}$ is $100 \times 11$ (1 intercept + 1 continuous + 9 dummy columns). *Want $p$ small relative to $n$ — helps ensure $\mathbf{X}'\mathbf{X}$ is invertible.*

*Discrete predictor taking values 1–100: continuous or categorical? Ask: can the levels be ordered? Is the difference between adjacent levels always the same (e.g. $2-1 \stackrel{?}{=} 100-99$)? If yes to both, treat as continuous.*

## Estimation via Least Squares

**Definition - Residual Sum of Squares**

$$RSS(\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} \hat{\epsilon}_i^2 = \sum_{i=1}^{n} \left(Y_i - \mathbf{x}_i'\hat{\boldsymbol{\beta}}\right)^2 = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = (\mathbf{Y} - \hat{\mathbf{Y}})'(\mathbf{Y} - \hat{\mathbf{Y}})$$

Same LS procedure as SLR: (1) set up the estimating equation, (2) differentiate w.r.t. $\boldsymbol{\beta}$ and set to $\mathbf{0}$, (3) solve for $\hat{\boldsymbol{\beta}}$, (4) unique minimizer iff $\mathbf{X}'\mathbf{X}$ is invertible (*full rank*) — see *multicollinearity*.

*Matrix calculus (Rencher & Schaalje §2.14): for constants $\mathbf{a}$, $\partial(\mathbf{a}'\mathbf{x})/\partial\mathbf{x} = \mathbf{a}$; for symmetric $\mathbf{A}$, $\partial(\mathbf{x}'\mathbf{A}\mathbf{x})/\partial\mathbf{x} = 2\mathbf{A}\mathbf{x}$; and $(\mathbf{A}\mathbf{B})' = \mathbf{B}'\mathbf{A}'$.*

**Proposition 8 LS Estimator.**
Assuming $(\mathbf{X}'\mathbf{X})^{-1}$ exists,

$$\boxed{\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}}$$

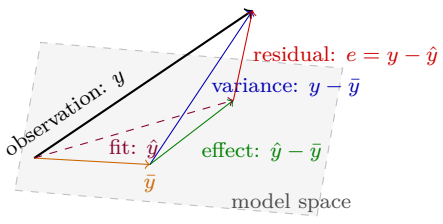
In practice, component matrices are easy to build from sums; only the inverse needs software.

$$\mathbf{X}'\mathbf{X} = \begin{pmatrix} n & \sum x_{i1} & \cdots & \sum x_{ip} \\ \sum x_{i1} & \sum x_{i1}^2 & \cdots & \sum x_{i1}x_{ip} \\ \vdots & \vdots & \ddots & \vdots \\ \sum x_{ip} & \sum x_{i1}x_{ip} & \cdots & \sum x_{ip}^2 \end{pmatrix}, \qquad \mathbf{X}'\mathbf{Y} = \begin{pmatrix} \sum y_i \\ \sum x_{i1}y_i \\ \vdots \\ \sum x_{ip}y_i \end{pmatrix}$$

**Definition - Hat Matrix**

$\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} = \mathbf{H}\mathbf{Y}$, where $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$ is the **hat/projection/influence** matrix: *idempotent* ($\mathbf{H}^2 = \mathbf{H}$), and projects $\mathbf{Y}$ onto $\hat{\mathbf{Y}}$ (*model space*).

- Diagonal entries $h_{ii} = \mathbf{x}_i'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_i$ are **leverages**: high leverage $\Rightarrow$ *outlier* in $X$, potential high influence on $\hat{\boldsymbol{\beta}}$ if deleted.
- Residuals: $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y} = \mathbf{M}\mathbf{Y}$, where $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$ is the **residual maker** ($\hat{\boldsymbol{\epsilon}} = \mathbf{M}\mathbf{Y}$) or **annihilator** matrix ($\mathbf{M}\mathbf{X} = \mathbf{0}$).




*$\hat{y}$ is the orthogonal projection of $y$ onto the model space; $e \perp$ model space.*

## Application Example

**Example 9 `mtcars` Dataset**
32 automobiles (1973–74 models); using `mpg` (response), `disp`, `hp`, `wt` (predictors):

$$mpg_i = \beta_0 + \beta_1 disp_i + \beta_2 hp_i + \beta_3 wt_i + \epsilon_i$$

Fitted: $\widehat{mpg} = 37 - 0.009\, disp - 0.03\, hp - 3.8\, wt$. *`disp` not significant ($p = 0.93$); `hp`, `wt` significant ($p = 0.011,\ 0.001$); $R^2 = 0.827$.*

**Example 10 Rencher & Schaalje Ex. 7.2 — Worked by Hand**
12 observations of $Y$ on $X_1, X_2$ give $\sum y_i = 90$, $\sum x_{i1} = 52$, $\sum x_{i2} = 102$, $\sum x_{i1}^2 = 296$, $\sum x_{i2}^2 = 1004$, $\sum x_{i1}x_{i2} = 536$, $\sum x_{i1}y_i = 482$, $\sum x_{i2}y_i = 872$, so

$$\mathbf{X'X} = \begin{pmatrix} 12 & 52 & 102 \\ 52 & 296 & 536 \\ 102 & 536 & 1004 \end{pmatrix}, \qquad \mathbf{X'Y} = \begin{pmatrix} 90 \\ 482 \\ 872 \end{pmatrix}$$

Inverting (software) and multiplying:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X'X})^{-1}\mathbf{X'Y} \approx \begin{pmatrix} 5.375 \\ 3.012 \\ -1.285 \end{pmatrix} \Rightarrow\ \hat{y}_i = 5.375 + 3.012x_{i1} - 1.285x_{i2}$$

**Example 11 Prediction**
Using Ex. 10 at $X_1 = 5$, $X_2 = 9$:

$$\hat{y} = 5.375 + 3.012(5) - 1.285(9) = 8.87$$

Interpretation: the mean response when $X_1 = 5, X_2 = 9$ is 8.87.

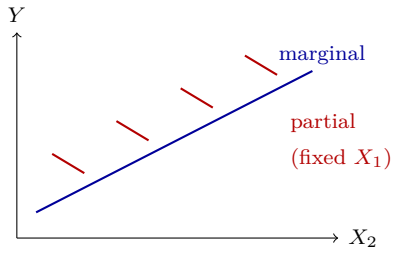
## Interpreting Coefficients

**Definition - Coefficient Interpretation**

$\hat{Y} = \hat{E}[Y \mid \mathbf{X} = \mathbf{x}] = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \cdots + \hat{\beta}_p x_p = \mathbf{x}'\hat{\boldsymbol{\beta}}$.

- $\hat{\beta}_0$: estimated avg. $Y$ when *all* predictors are 0.
- $\hat{\beta}_j$ $(j \geq 1)$: estimated avg. change in $Y$ per 1-unit increase in $x_j$, **holding all other predictors fixed**.

*Remark* (Conditional Nature of MLR). Coefficients are only valid *holding other predictors fixed* — an SLR fit on one predictor alone can differ drastically (even in sign) from its MLR coefficient. In Ex. 10, SLR gives $\hat{y} = 1.86 + 1.30x_1$ and $\hat{y} = 0.86 + 0.78x_2$ (both *positive* slopes), but jointly $\hat{\beta}_2 = -1.285$ (*negative*) — because MLR's $\hat{\beta}_2$ is the effect of $X_2$ *within* a fixed value of $X_1$ (a *partial regression*), not *marginally*. *Ignoring a correlated predictor can flip the apparent sign of an effect.*




*Marginal slope (blue, ignoring $X_1$) vs. partial slopes (red, at fixed $X_1$) — opposite sign here.*

**Example 12 Interpreting an Indicator Variable**
Add `am` ($\mathbb{I}(am = 1)$ = manual) to the `mtcars` model:

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 disp + \hat{\beta}_2 hp + \hat{\beta}_3 wt + \hat{\beta}_4\, \mathbb{I}(am = 1), \qquad \hat{\beta}_4 = 2.16$$

$\hat{\beta}_4$ = estimated avg. difference in *mpg* between manual and automatic transmission, holding `disp`, `hp`, `wt` fixed: the intercept is $\hat{\beta}_0$ when $am = 0$, and $\hat{\beta}_0 + \hat{\beta}_4$ when $am = 1$ — a parallel vertical shift, same slopes.

**Definition - Interaction**

An **interaction effect** exists when the joint effect of two (or more) predictors differs from their separate effects; include it as a product term $x_j x_k$ (works for any combination of numeric/categorical predictors). *E.g. is the smoking–lung-cancer association the same in rural vs. urban areas? Does a renovation raise price in the suburbs but not downtown? Does drug dosage increase response in females but decrease it in males?*

*"Linear" still just means linear in $\boldsymbol{\beta}$: e.g. $Y = \beta_0 + \beta_1 x_1 x_2 + \beta_2 \sin(x_2) + \beta_3 x_3^2$ is a linear model.*

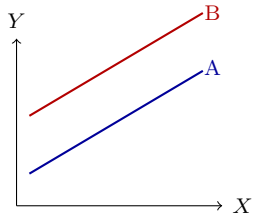
**Example 13 Interaction — `mtcars`**

$$\widehat{mpg} = \hat{\beta}_0 + \hat{\beta}_1 wt + \hat{\beta}_2 \mathbb{I}(am = 1) + \hat{\beta}_3 (wt \times \mathbb{I}(am = 1))$$

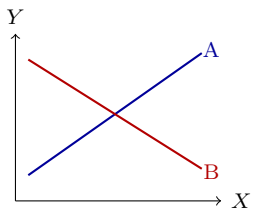
Fitted: $31.42 - 3.79wt + 14.88\,\mathbb{I}(am = 1) - 5.30(wt \times \mathbb{I}(am = 1))$. Slope of `wt` is $-3.79$ for automatic ($am = 0$); for manual ($am = 1$) it is 5.30 *more negative*, i.e. $-9.09$ — different intercept *and* slope per group.

*Remark* (Detecting Interaction Visually). No interaction ⇒ fitted lines/planes for each category are **parallel**. **Non-parallel** slopes (e.g. `mpg` vs. `wt` across `cyl` $\in \{4, 6, 8\}$) suggest an interaction (formal test later). With $> 2$ predictors, "lines" become planes/hyperplanes.

*Plotting caution: don't lead the viewer by extrapolating fitted lines beyond each group's observed range — and don't recode a numeric predictor (e.g. `cyl`) as categorical for a plot like this unless there's a substantive reason to.*




*No interaction: parallel slopes*




*Interaction: non-parallel (crossing) slopes*

*A categorical predictor with $c > 2$ levels still only needs $c - 1$ dummy columns, generated automatically by software (e.g.* `factor(cyl)` *for* `cyl` $\in \{4, 6, 8\}$ *creates 2 columns, baseline = smallest level, 4).*

**Example 14 Try at Home**

Given the data below:

| $Y$ | $X_1$ | $X_2$ |
|---|---|---|
| $-0.2$ | 0.5 | 0.5 |
| $-1.1$ | 0.8 | 0.7 |
| $-0.8$ | 0.6 | $-0.5$ |
| $-0.7$ | $-0.2$ | $-0.5$ |

Write out $\mathbf{X}$ (with intercept column) and compute $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ by hand; check against `lm()`.

# Week 3: Assumptions of Linear Regression Models

| **Learning Objectives** | |
|---|---|
| *Basic* | State the 4 linear regression assumptions and what each means §; use residual plots and QQ plots to check them §. |
| *Adv.* | Explain why residuals have non-constant variance and when to standardize; check the two extra conditions needed before residual plots can be trusted in MLR §. |

## Review: MLR, Indicators and Interactions

*Remark* (MLR in Matrix Form). $Y_i = \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} + \epsilon_i = \mathbf{x}_i'\boldsymbol{\beta} + \epsilon_i$, $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$, i.e. $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$. $\mathbf{Y}$ *= column vector of responses;* $\mathbf{X}$ *= design matrix;* $N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$ *= multivariate normal.*

**Example 15 Design Matrix with Interaction — `mtcars`**
`lm(mpg ~ wt * factor(cyl))` fits

$$mpg = \beta_0 + \beta_1 wt + \underbrace{\beta_2 \mathbb{I}(cyl = 6) + \beta_3 \mathbb{I}(cyl = 8)}_{\text{main effect for cyl}} + \underbrace{\beta_4 wt \cdot \mathbb{I}(cyl = 6) + \beta_5 wt \cdot \mathbb{I}(cyl = 8)}_{\text{interaction wt}\times\text{cyl}} + \epsilon$$

`model.matrix()` has columns 1, $X_1 = wt$, $X_2 = \mathbb{I}(cyl = 6)$, $X_3 = \mathbb{I}(cyl = 8)$, $X_4 = X_1X_2$, $X_5 = X_1X_3$. *Main effect ⇒ different* intercepts *per group; interaction ⇒ different* slopes *per group.* Fitted: $\hat{\beta}_0 = 39.571$, $\hat{\beta}_1 = -5.647$, $\hat{\beta}_2 = -11.162$, $\hat{\beta}_3 = -15.703$, $\hat{\beta}_4 = 2.867$, $\hat{\beta}_5 = 3.455$ ($R^2 = 0.8616$).

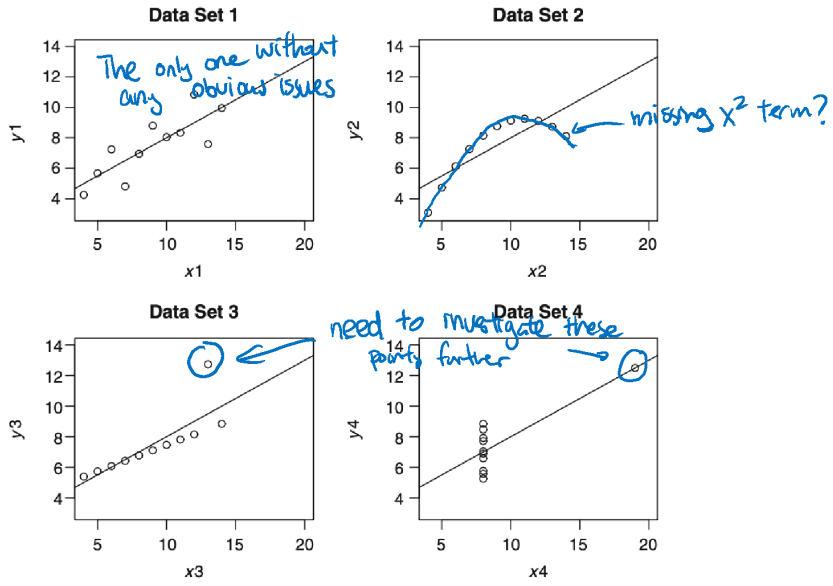
## Linear Regression Assumptions

*Remark* (Why We Need Assumptions). $Y_1, \ldots, Y_n$ are random; treat $\mathbf{X}_1, \ldots, \mathbf{X}_n$ as fixed. So $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is random, with a **sampling distribution** (like $\hat{\mu}$ in 2nd year). We need assumptions to determine that sampling distribution, to have a valid model fit, and to make statistical inferences.

**Example 16 Anscombe's Quartet**
Anscombe (1973) built 4 datasets (in R: `anscombe`) with nearly identical summary statistics:

$$\bar{x} = 9, \quad \bar{y} \approx 7.5, \quad \text{Var}(x) = 11, \quad \text{Var}(y) \approx 4.13, \quad \text{corr}(x, y) \approx 0.82, \quad \hat{y} = 3 + 0.5x$$

for *all four*, yet they look completely different when plotted. *Always graph your data! See Sheather §3.1 (pg 45).*




*Set 1: the only one without obvious issues. Set 2: missing an $x^2$ term? Sets 3 & 4: need to investigate the circled points further.*

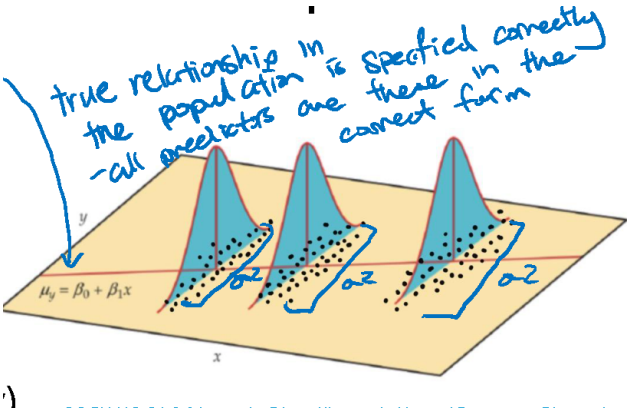
*Remark.* "All models are wrong, but some are useful" — George Box (1976).

**Definition - The 4 Assumptions**

Assumptions are about the **population**, not the sample, and we implicitly make them **every** time we fit a model.

1. **Linearity** (a.k.a. **mean zero errors**): $E(\boldsymbol{\epsilon} \mid \mathbf{X}) = \mathbf{0}$, i.e. $E(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$
2. **Uncorrelated errors** (a.k.a. independence, no autocorrelation, no serial correlation): $\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(y_i, y_j) = 0$ for $i \neq j$
3. **Constant error variance** (**homoskedasticity**): $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$, i.e. $\text{Var}(\epsilon_i \mid X) = \sigma^2$
4. **Normal errors**: $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$, i.e. $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

*The single statement $\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$ implies all 4 assumptions. Normality is technically not needed for OLS, but is needed for inference on $\boldsymbol{\beta}$.*




*Each conditional distribution of $Y$ is normal, centred on the line, with the same spread $\sigma^2$.*

### What Each Assumption Means

- **Linearity / mean zero errors**: the true relationship is linear in the *coefficients*, and is exactly $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with
  - no predictors omitted from $\mathbf{X}$ that should be present,
  - no predictors included in $\mathbf{X}$ that should not be present,
  - no predictors in $\mathbf{X}$ in the wrong functional form *(x is transformed if necessary)*.

  Violations: omitting a predictor known to influence the response; fitting a linear model when the truth is $y_i = \log(\beta_0 + \beta_1 x_i + \epsilon_i)$; including $x$ when it should be $x + x^2$. *U-shaped data: $Y = \beta_0 + \beta_1 x$ violates linearity, but $Y = \beta_0 + \beta_1 x + \beta_2 x^2$ is fine — still linear in $\boldsymbol{\beta}$.*
- **Uncorrelated errors**: no data point in the population is related to any other (knowing one tells you nothing about another). Depends on *how samples are taken*. Violations: stock prices, weather data, repeated measurements on the same person, family/social groups.
- **Constant variance**: each conditional distribution has the same *spread*; the only difference between conditional distributions of $Y_i$ is that the mean shifts.
- **Normality**: each conditional distribution has the same (normal) *shape*. Harder to verify in small samples — *but also more important in small samples*. **Not** needed to estimate coefficients by least squares (the normal distribution is never used in that derivation).

### Importance of Assumptions

- Linearity ⇒ coefficients are estimated **unbiasedly**
- Uncorrelated errors ⇒ correct **precision** of estimates
- Constant variance ⇒ reasonable estimates of variability for all conditional means
- Normality ⇒ can use properties of normal RVs for **inference** (e.g. CIs)
- Nothing stops us from fitting an invalid model with violated assumptions!

## Verifying Assumptions with Residual Plots

### Definition - Residuals

Residuals $\hat{\epsilon}$ are the estimated errors (distance from the fitted line). $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$

are our (sample) estimates of the errors, with

$$E[\hat{\epsilon}_i \mid X] = 0, \qquad \text{Var}[\hat{\epsilon}_i \mid X] = \sigma^2(1 - h_{ii}),$$

where $h_{ii}$ is the $i$-th diagonal entry of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$. *The errors all have the same variance, but the residuals' variances depend on $X$.* Estimate $\sigma^2$ by

$$\hat{\sigma}^2 = s^2 = \frac{1}{n-(p+1)} \sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2 = \frac{1}{n-(p+1)} \sum_{i=1}^{n} \hat{\epsilon}_i^2$$

*$df = n - p - 1$: $n$ observations minus $p+1$ $\beta$'s. The sum is the same as in SLR.*

**Definition - Standardized Residuals**

Since residuals don't all have the same variance, standardize:

$$r_i = \frac{\hat{\epsilon}_i}{\hat{\sigma}\sqrt{1 - h_{ii}}}$$

**Proposition 17** $0 \leq h_{ii} \leq 1$.
$\mathbf{H}$ is a projection matrix, so $\mathbf{H} = \mathbf{H}^2$, giving

$$h_{ii} = h_{ii}^2 + \sum_{j \neq i} h_{ij}^2 \;\Rightarrow\; h_{ii} \geq h_{ii}^2 \;\Rightarrow\; 0 \leq h_{ii} \leq 1.$$

If $h_{ii} \approx 1$ (extremely high leverage), $\hat{\epsilon}_i$ has variance $\approx 0$, so it is forced close to 0. *Most $h_{ii}$ will be small (close to 0).* Standardized residual plots are more informative when high-leverage points exist. *Often, regular residual plots are used.*

*Remark* (Why Residuals Reveal Violations). With population errors $\epsilon_i = y_i - E(y_i \mid x_i)$ and residuals $\hat{\epsilon}_i = y_i - \hat{E}(y_i \mid x_i)$: residuals capture the noise left over after estimating the trend. If $\hat{\boldsymbol{\beta}} \approx \boldsymbol{\beta}$,

$$\underbrace{\hat{\boldsymbol{\epsilon}}}_{\text{observed}} = \mathbf{Y} - \hat{\mathbf{Y}} = \mathbf{X}(\boldsymbol{\beta} - \hat{\boldsymbol{\beta}}) + \boldsymbol{\epsilon} \approx \underbrace{\boldsymbol{\epsilon}}_{\text{unobserved}}$$

So violations show up in the residuals. E.g. if the truth is $y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i1}^2 + \epsilon_i$ but we fit $\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_{i1}$:

$$\hat{\epsilon}_i = (\beta_0 - \hat{\beta}_0) + (\beta_1 - \hat{\beta}_1)x_{i1} + \beta_2 x_{i1}^2 + \epsilon_i \approx \beta_2 x_{i1}^2 + \epsilon_i$$

*Residuals show what was missed in your specified model.*

**What Good Residuals Look Like**

- Uncorrelated with the fitted values and predictors *(no patterns)*
- Residuals vs. fitted values or any predictor have **constant** mean 0
- **No systematic patterns** — *should look like a random cloud*
- Roughly normally distributed *A funnel (spread growing with $x$ or $\hat{y}$) violates constant variance.*

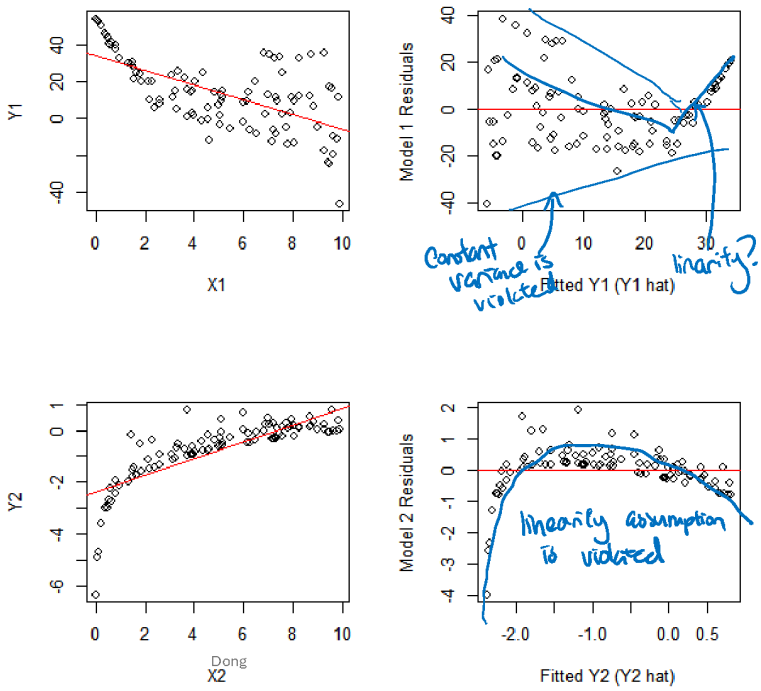
*Remark* (No Perfect Diagnostic). Diagnostic plots may not capture violations, and may be misleading (especially for small $n$; more trustworthy for large $n$). A violation of one assumption may affect the diagnostic plot for *another* assumption.

| Plot | Checks |
|---|---|
| Residuals vs. fitted values (scatter) | linearity, uncorrelated errors, constant variance |
| Residuals vs. each predictor (scatter) | linearity, uncorrelated errors, constant variance |
| Histogram of (standardized) residuals | normality, mean zero |
| Normal QQ plot of residuals | normality |

*Any of these can use standardized residuals instead.*

**Example 18 You Try**

Scatterplots with line of best fit (left) and residuals vs. fitted (right) for two datasets. What patterns do we see?




$(X_1, Y_1)$: the residuals fan out, so **constant variance is violated**; the curve on the right also suggests a linearity problem. $(X_2, Y_2)$: the residuals follow an arch, so **linearity is violated**.

**Definition - Normal QQ Plot**

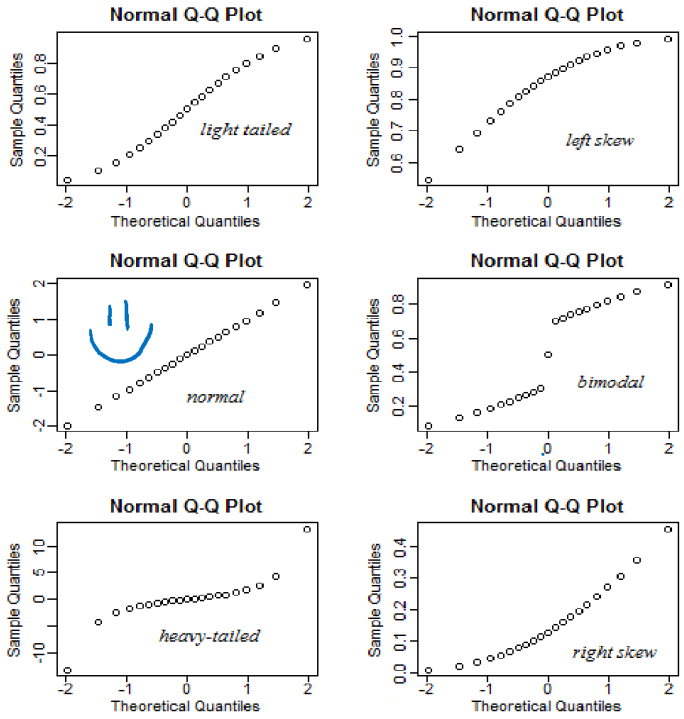
To check normality with a QQ plot of the $r_i$:

1. Sort the data from smallest to largest; record the percentile each value occupies.
2. Find the standard normal quantile ($z$-score) at the same percentiles.
3. Plot each data point $x_i$ against its $z_i$.

If the distribution is close to normal, the points form a **straight line**. In R: `qqnorm(model$residuals)` and `qqline(model$residuals)`.

*Remark* (Reading QQ Plots). Typical deviations from a straight line:

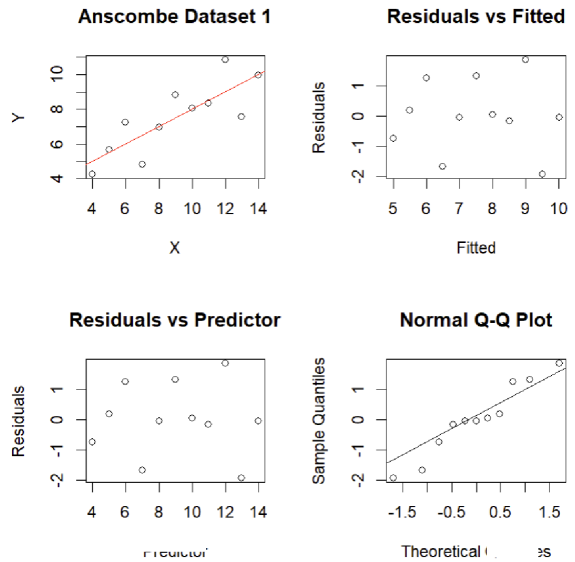
- **Outliers**: odd values far from the rest
- **Curvature at both ends**: long (heavy) or short (light) tails
- **Convex/concave curvature**: asymmetry/skewness
- **Horizontal segments, plateaus, gaps** (e.g. bimodal)



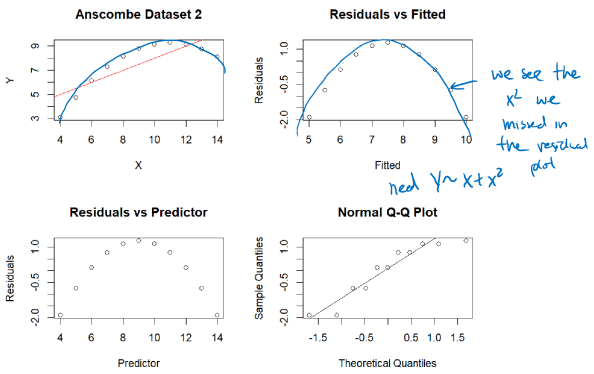

**Example 19 Back to Anscombe's Quartet**

For each dataset: scatter with fit, residuals vs. fitted, residuals vs. predictor, normal QQ plot.

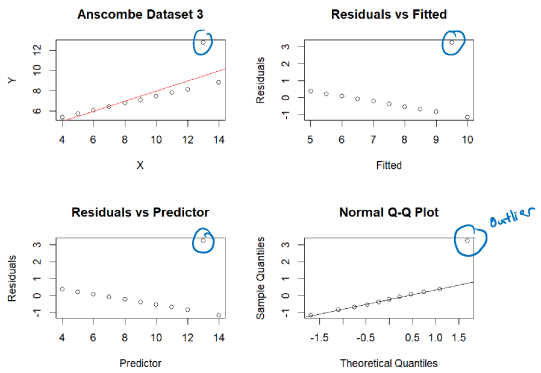
```
model1 <- lm(y1 ~ x1, data = anscombe)
y_hat <- fitted(model1); e_hat <- resid(model1)
par(mfrow = c(2, 2))   # 2x2 grid of plots
plot(anscombe$x1, anscombe$y1); abline(model1, col = "red")
plot(y_hat, e_hat)            # residuals vs fitted
plot(anscombe$x1, e_hat)      # residuals vs predictor
qqnorm(e_hat); qqline(e_hat)
```



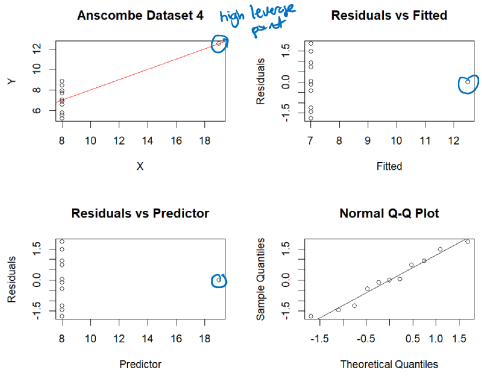

*Set 1: random cloud, QQ roughly straight — no issues*




*Set 2: arch in the residuals*



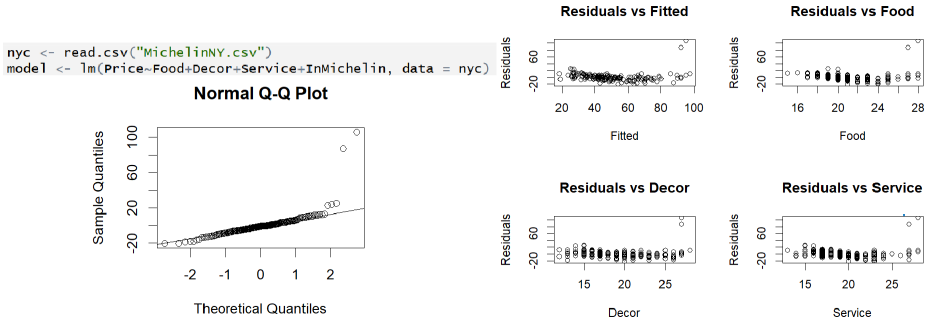

*Set 3: one outlier*




*Set 4: one high-leverage point*

*Set 2: we see the $x^2$ we missed in the residual plot — need $Y \sim X + X^2$. Set 3: outlier stands out in every plot. Set 4: the high-leverage point at $x = 19$ has residual $\approx 0$ (since $h_{ii} \approx 1$).*

*Remark* (Discrete Predictors). Random vertical "bands" of residuals indicate **no** violation when predictors are discrete (e.g. Michelin NY data, `lm(Price ~ Food + Decor + Service + InMichelin)`).



### Explore and Understand the Data

- Do exploratory data analysis *before* fitting a model — plot plot plot!
  - Is most of your data concentrated in a small area? Are the response or predictors skewed? *Doesn't mean something is wrong, but look closer.*
- Existing literature informs knowledge of the true relationship: what are the variables measuring? Are transformations typically done? What predictors mattered in prior studies? How should predictors be included?

**Example 20 Think About the Data Collection Process**
Which assumptions may be violated?

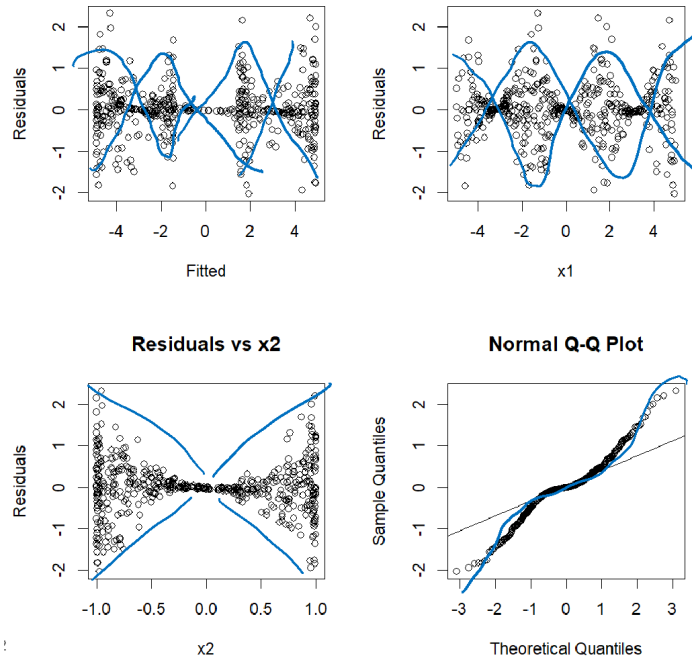
- "Data was collected by voluntary response survey..."
  Linearity: there could be response bias that isn't accounted for in the model.
- "Each variable was measured every day for a month..."
  Uncorrelated errors (consecutive days are related).
- "Neighbourhoods were randomly sampled and all subjects from selected neighbourhoods were included..."
  Uncorrelated errors (people in the same neighbourhood are related).

Some violation is unavoidable (all models are wrong!) — consider the context of the question and the cost of any errors.

## Additional Conditions for Multiple Regression Models

*The instructor crossed out "Linear" in "Multiple Linear Models": these conditions apply to multiple regression models in general.*

**Example 21 What's Wrong Here?**
The model $Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2$ was fit, giving these residual plots:

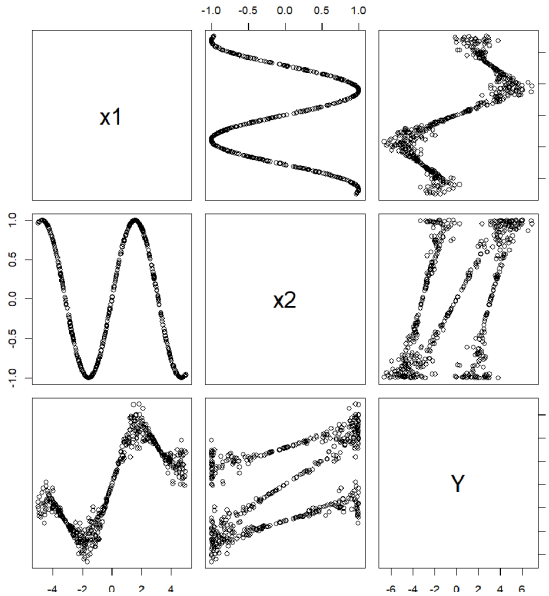



The plots seem to suggest many violations (curved patterns, funnels, non-normal QQ). But the data were generated as

```
x1 = runif(n, -5, 5)
x2 = sin(x1)
Y  = x1 + 3*x2 + rnorm(n, 0, x2^2)
```

The mean is correctly specified and the errors are normal and independent, so **non-constant variance** (sd $= x_2^2$) is the *only* violated assumption.

**Example 22 Why Do We See More Issues Than the Funnel?**



The pairs plot shows $x_2 = \sin(x_1)$: the predictors have a strongly *non-linear* relationship. That makes the residual plots show patterns that are artifacts of the predictor relationship, not actual extra violations.

**Definition - Conditions for Interpreting Residual Plots in MLR**

Coefficients are interpreted holding all other predictors fixed (conditioning on them). Before looking at residual plots, check (Li & Duan, 1989; Sheather pg 155):

1. **Conditional mean response condition**: the relationship between the response (or transformed response) and the predictors is linear,

$$E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}) = g(\beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p)$$

2. **Conditional mean predictor condition**: each pair of predictors is linearly related,

$$E(X_i \mid X_j) = \alpha_0 + \alpha_1 X_j$$

*These are for interpreting residual plots.*

*Remark* (Why These Conditions?).

- More predictors = more complexity. Residual plots may show patterns that are artifacts of relationships between predictors, not actual violations, and the pattern seen may not be what is actually missing or wrong.
- Recall: violation of one assumption may affect the diagnostic plot for a separate assumption.
- Complex relationships among predictors (or between predictors and response) make residual plots unreliable. Without checking both conditions, we **cannot** use residual plots to identify *specific* violations.

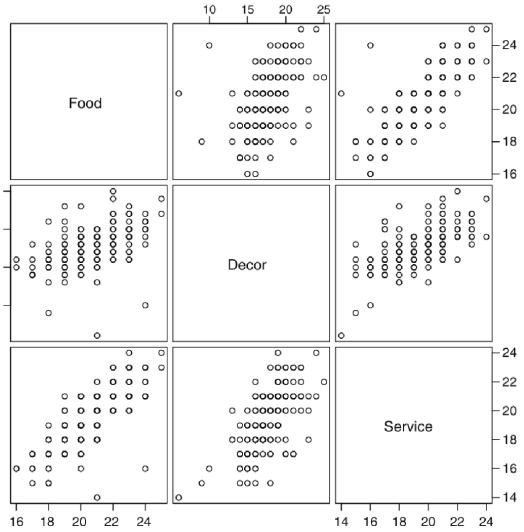
**How to Check**

- **Response vs. fitted values**: identifies a non-linear relationship. Look for deviations from the line $\hat{Y} = Y$.
- **Pairs plot** (scatterplot matrix): identifies pairwise relationships between predictors. Look for curves and non-linear patterns.

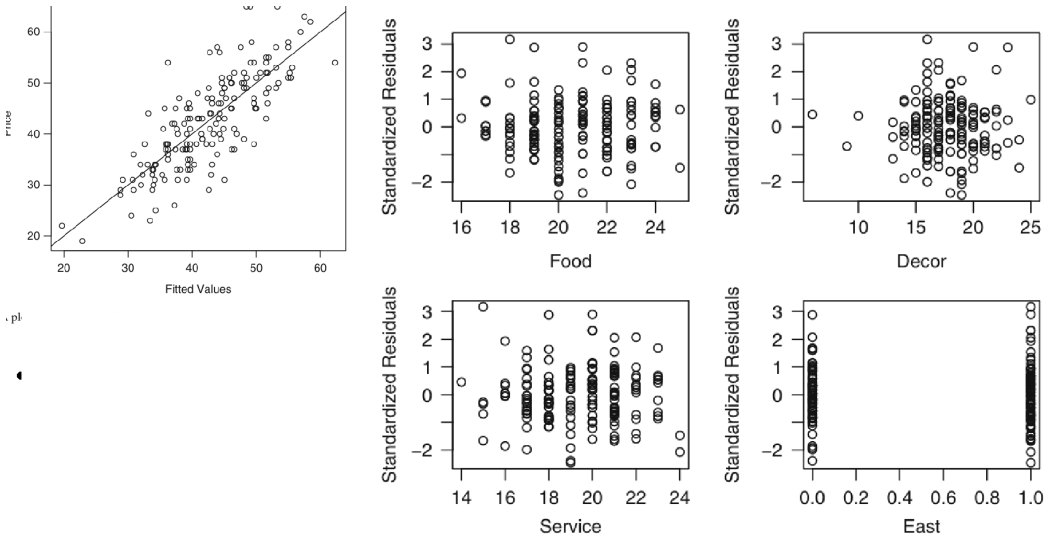
*Cook & Weisberg (1999): "Using residuals to guide model development will often result in misdirection, or at best more work than would otherwise be necessary."*

**Example 23 Both Conditions Hold — NYC Italian Restaurants**

3 numeric predictors (Food, Decor, Service) plus East (indicator).




The predictors are (approximately) linearly related ⇒ **conditional mean predictor condition holds**.



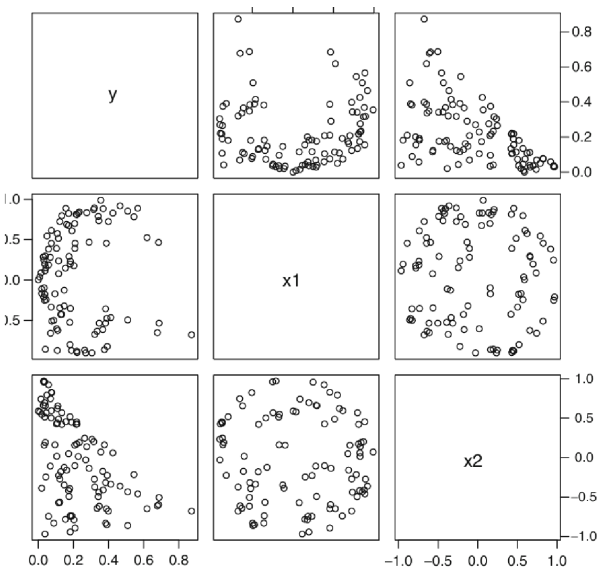

**ure 6.2** Plots of standardized residuals against each predictor variable

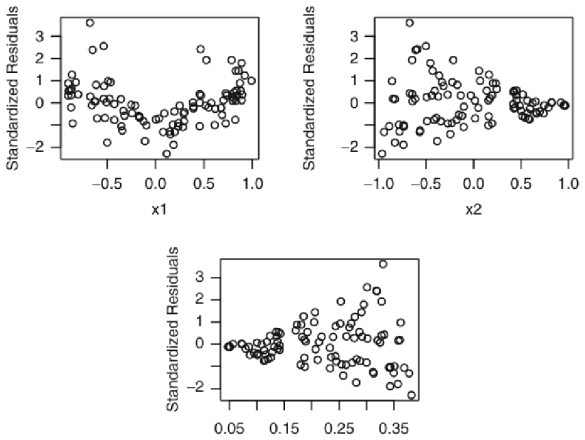
Price vs. fitted values follows the line ⇒ **conditional mean response condition holds** with $g(z) = z$, so the standardized residual plots can be trusted (and look fine).

**Example 24 Conditional Mean Response Fails — `alr4::caution`**
(Sheather pg 159.) $x_1, x_2$ generated on the unit circle, errors i.i.d. $N(0, 1)$, and

$$E(Y \mid X) = \frac{|x_1|}{2 + (1.5 + x_2)^2} = \frac{g_1(x_1)}{g_2(x_2)},$$

but the model fit is $Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + e$.




- $y$ has a complicated relationship with $x_1$ and $x_2$ $\Rightarrow$ conditional mean response condition **violated**.
- Not enough visual evidence that $x_1$ vs. $x_2$ is non-linear $\Rightarrow$ conditional mean predictor condition holds.
- The residual plots look like a constant-variance problem, but we *know* the errors are i.i.d. $N(0, 1)$.

**We can't say which part of the model is misspecified from the residual plots.** All we can say is that the model fit is invalid.

## Summary

- Choose which predictors to use (formal variable selection later)
- Fit the model
- Always plot (extensively!) to check whether the assumptions are valid

*Remark* (Module Takeaways).

- What are the assumptions of linear regression and what do they mean?
- What tools assess whether each assumption holds, and what do we look for?
- Why check the additional conditions in multiple regression models?
- What aspects of the data, data collection or population suggest an assumption may not hold?

*Next week: correcting and mitigating violated assumptions.*

# Week 4: Mitigating Violated Assumptions

## Review: Assumptions and Residual Plots

*Remark* (The 4 Assumptions (Week 3)). Assumptions are about the **population**, not the sample, and are made implicitly **every** time we fit a model. All four are packed into

$$\boldsymbol{\epsilon} \mid \mathbf{X} \sim N_n(\underbrace{\mathbf{0}}_{\text{linearity}}, \underbrace{\sigma^2}_{\text{const. var.}} \underbrace{\mathbf{I}}_{\text{uncorr.}}) \quad \text{or} \quad \mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}),$$

with the $N$ itself being the normality assumption.

1. **Linearity** (mean zero errors): $E(\boldsymbol{\epsilon} \mid \mathbf{X}) = \mathbf{0}$, i.e. $E(\mathbf{Y} \mid \mathbf{X}) = \mathbf{X}\boldsymbol{\beta}$
2. **Uncorrelated errors**: $\text{Cov}(\epsilon_i, \epsilon_j) = \text{Cov}(y_i, y_j) = 0$
3. **Constant variance**: $\text{Var}(\boldsymbol{\epsilon} \mid \mathbf{X}) = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$
4. **Normal errors**: $\epsilon_i \overset{iid}{\sim} N(0, \sigma^2)$

*1–3 are needed for OLS. Normality is not needed for OLS, but is needed for inference.*

### Spotting Violations with Residuals

- **Linearity**: any systematic pattern, especially curves or other functions of predictors.
- **Uncorrelated errors**: large clusters of many points, or a pattern in **time** or other sequencing information. *Check how the data was collected.*
- **Constant variance**: any systematic pattern, especially a fanning/funnel pattern with increasing or decreasing spread.
- **Normality**: QQ plot — look for stark deviations/curving/wiggling from the diagonal line *(minimal deviations ok)*.

## Introduction to Transformations

### Definition - Transformed Model

The main reasons to transform are to overcome **non-linearity** and **non-constant variance**. We fit

$$g_Y(Y) = \beta_0 + \beta_1 g_{X_1}(X_1) + \beta_2 g_{X_2}(X_2) + \cdots + \epsilon$$

where the $g(\cdot)$ are transformations applied to the original data.

- We can transform any subset of the response and predictors.
- We can only interpret the model in terms of the *transformed* predictors/response.

### Choosing a Transformation

1. Plot the response vs. the predictors (or residuals) to look for an appropriate transformation.
2. Choose a transformation to try to fix the residual plot.
3. Plot again to see if the issue was fixed.

**Drawback**: may make interpretation of the estimated coefficients difficult. **Advantage**: often improves prediction and inference.

**Definition - Common Transformations**

- **Logarithmic**: $g(z) = \log(z)$ *(usually natural log; make sure the variable is strictly positive)*
- **Power**: $g(z) = z^\lambda$ for $\lambda > 0$
- **Inverse power**: $g(z) = 1/z^\lambda$ for $\lambda > 0$

*$\lambda$ is usually a "nice" number.* These can be applied to predictors or the response; e.g. from SLR $Y = \beta_0 + \beta_1 X + \epsilon$:

$$Y = \beta_0 + \beta_1 X^2 + \epsilon, \qquad \log(Y) = \beta_0 + \beta_1 X + \epsilon, \qquad \sqrt{Y} = \beta_0 + \beta_1 \exp(X) + \epsilon$$

*The last two are no longer linear in $Y$.*

*Remark* (Transformations for Skewed Data). Instead of $X$ use $X^* = f(X)$ (same for $Y$); any numerical variable can be transformed.

- **Right skew**: $\log(x - c)$, $\sqrt{x - c}$, etc.
- **Left skew**: $x^2$, $\sqrt{c - x}$, $\log(c - x)$, etc.
- Check graphically whether it improved.

There are analytic methods to help choose a transformation (Box–Cox, below); these common ones are often the recommended options.

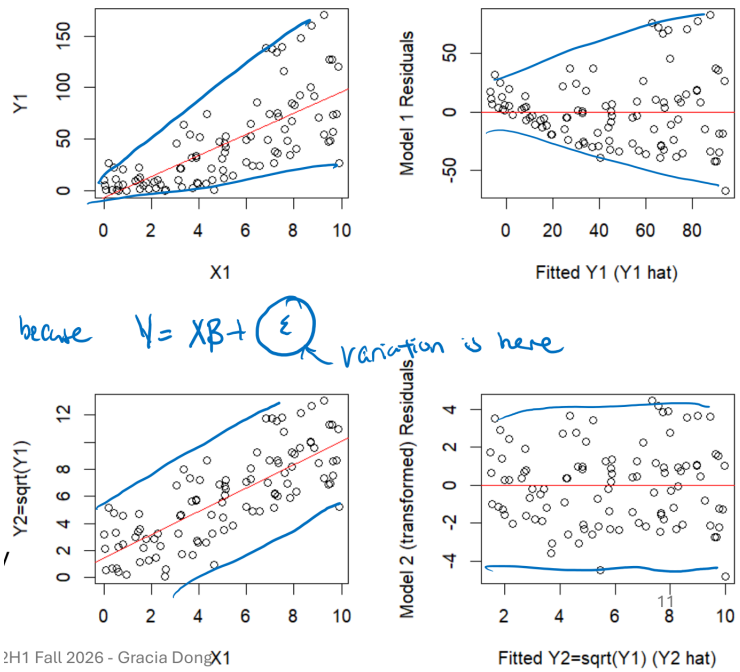
### Variance Stabilizing Transformations

**Definition - Variance Stabilizing Transformation (VST)**

A transformation that specifically targets violation of **constant variance**. It is applied to the **response $Y$ only** — *because in $Y = X\beta + \epsilon$, the variation is in $\epsilon$; there is no randomness associated with $X$.* Best practice: select transformations that target the violation you observe.

The choice depends on the exact situation and data:

- Identifiable data type (e.g. count data, binary data) or specific variable context
- Experience from similar contexts or data types
- No one transformation works for every model — trial and error




*Top: $Y_1$ vs. $X_1$ and residuals fan out. Bottom: after $Y_2 = \sqrt{Y_1}$, the residual band is roughly constant.*

**Proposition 25 Delta Method Approximation.**
Let $f(Y)$ be the transformation. First-order Taylor expansion around the mean:

$$f(Y) = f\big(E(Y)\big) + f'\big(E(Y)\big)\big(Y - E(Y)\big) + \cdots$$

Take variance of both sides. $f(E(Y))$, $f'(E(Y))$ and $E(Y)$ are *constants*, so $\mathrm{Var}(f(E(Y))) = 0$ and the covariance term is $\mathrm{Cov}(\text{constant}, \text{random}) = 0$, leaving

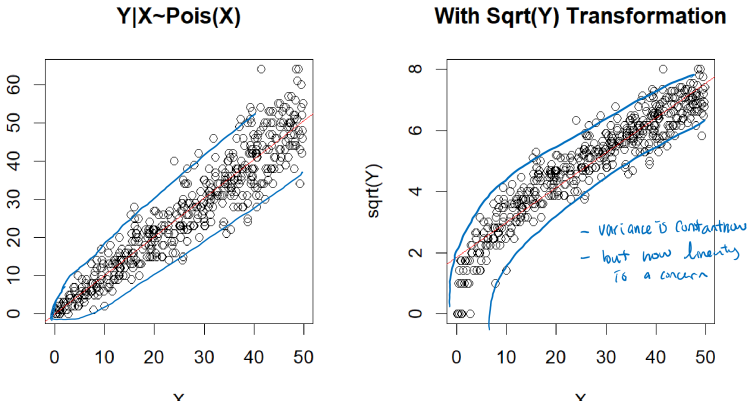
$$\boxed{\mathrm{Var}\left(f(Y)\right) \approx \left[f'(E(Y))\right]^2 \mathrm{Var}(Y)}$$

The correct choice of $f$ cancels out the non-constant part of $\mathrm{Var}(Y)$.

**Example 26 Poisson Response**
Suppose $Y \mid X \sim \mathrm{Poi}(X)$, so $E(Y \mid X) = \mathrm{Var}(Y \mid X) = X$. *This also violates normality, but focus on non-constant variance for now.* Is $f(Y) = \sqrt{Y}$ variance-stabilizing?
$f(y) = y^{1/2} \Rightarrow f'(y) = \frac{1}{2}y^{-1/2}$, so $f'(E(Y \mid X)) = \frac{1}{2}X^{-1/2}$. Then

$$\mathrm{Var}\left(f(Y) \mid X\right) \approx \left(\tfrac{1}{2}X^{-1/2}\right)^2 X = \tfrac{1}{4},$$

which is **constant**.




*After $\sqrt{Y}$ the variance is constant, but now linearity is a concern.*

### Notes on VSTs

- Only transform $Y$.
- Can potentially make other assumption violations worse.
- Pick based on what the plots look like and the direction of the funnel:
  - No obvious choice; common ones when variance increases with $Y$: $\ln(y-c)$, $\sqrt{y-c}$, etc.
  - If variance decreases with $Y$, flip to $\ln(c-y)$, ...
- Apply transformation, fit model, then plot again.

**Proposition 27 Common VSTs.**

| If $\mathrm{Var}(Y \mid X) \propto$ | use $f(Y) \propto$ |
|---|---|
| $E(Y \mid X)$ | $\sqrt{Y}$ *(could be $\sqrt{Y-c}$, $\sqrt{c-Y}$, ...)* |
| $E(Y \mid X)^2$ | $\log(Y)$ |
| $E(Y \mid X)^3$ | $1/\sqrt{Y}$ |
| $E(Y \mid X)^4$ | $1/Y$ |

It can be hard to tell visually what $\mathrm{Var}(Y \mid X)$ is proportional to — try different transformations and check for improvement.

**Exercise 28 Verify the Table.**
Verify each row using $\mathrm{Var}(f(Y) \mid X) \approx [f'(E(Y \mid X))]^2 \mathrm{Var}(Y \mid X)$.
Write $\mu = E(Y \mid X)$ and $\mathrm{Var}(Y \mid X) = k\mu^m$. log: $f' = 1/\mu$, so $\mu^{-2} \cdot k\mu^2 = k$. $1/\sqrt{Y}$: $f' = -\frac{1}{2}\mu^{-3/2}$, so $\frac{1}{4}\mu^{-3} \cdot k\mu^3 = k/4$. $1/Y$: $f' = -\mu^{-2}$, so $\mu^{-4} \cdot k\mu^4 = k$. All constant.

## Box–Cox Transformations for Normality

*Remark* (Motivation). Last week we eyeballed transformations for linearity from residual plots and scatterplots. Whatever "signal" the line of best fit misses is most likely a missing predictor (e.g. an arch in residuals vs. Distance ⇒ add Distance$^2$; a concave $\sqrt{Y}$ vs. $X$ ⇒ try $\sqrt{X}$). Sometimes the needed transformation isn't obvious even from residual vs. predictor plots, so we find the "best" one **analytically**.

**Definition - Box–Cox Transformation (Box & Cox, 1964)**

Power transformations, chosen to improve **normality** violations. Can be used on the response, on the predictor(s), or on both simultaneously. *Side effect: can help with linearity.* In this class we won't use it exactly as intended due to its complexity.
Further reading: Weisberg §8.1.2–8.1.3 (pg 188); derivation: Appendix A.12 (pg 312).

*Remark* (Maximum Likelihood). Box–Cox uses ML to estimate the power; the estimate gives the best approximation to normality among all powers. The normal log-likelihood in SLR (assumptions holding) is

$$\log L(\beta_0, \beta_1, \sigma^2 \mid Y) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log(\sigma^2) - \frac{1}{2\sigma^2}\underbrace{\sum (y_i - \beta_0 - \beta_1 x_i)^2}_{RSS}$$

*For MLR, just include more $\beta_j x_j$ in the last term.*

**Definition - How Box–Cox Works**

Modify the RSS, i.e. fit a new model with a different response:

$$RSS = \sum \left(\Psi_M(Y, \lambda) - \beta_0 - \beta_1 x_i\right)^2$$

$$\Psi_M(\mathbf{Y}, \lambda) = \mathrm{gm}(\mathbf{Y})^{1-\lambda}\, \Psi_S(y_i, \lambda) = \begin{cases} \left(e^{\frac{1}{n}\sum \log y_i}\right)^{1-\lambda} \dfrac{y_i^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \left(e^{\frac{1}{n}\sum \log y_i}\right)^{1-\lambda} \log(y_i), & \lambda = 0 \end{cases}$$

- $\mathrm{gm}(\mathbf{Y}) = e^{\frac{1}{n}\sum \log y_i}$ is the **geometric mean** of $Y$ *(roughly constant if $n$ is large).*
- $\Psi_S(y_i, \lambda) = \frac{y_i^\lambda - 1}{\lambda}$ or $\log(y_i)$ is a **s**caled power transformation.
- $\Psi_M(\mathbf{Y}, \lambda)$ is a **m**odified power transformation applied to $\mathbf{Y}$.

This is very complicated, so for interpretability we instead: get the ML estimate $\hat{\lambda}$ (in R), then just use $\mathbf{Y}^\lambda$ (or $\log \mathbf{Y}$).

*Remark* (Box–Cox on $X$, and on $X$ & $Y$). Predictors: $RSS = \sum \left(y_i - \beta_0 - \beta_1 \Psi_S(X, \lambda)\right)^2$, with $\Psi_S(X, \lambda) = \frac{X^\lambda - 1}{\lambda}$ $(\lambda \neq 0)$ or $\log X$ $(\lambda = 0)$.
Both simultaneously: $RSS = \sum \left(\Psi_M(Y, \lambda_y) - \beta_0 - \beta_1 \Psi_S(X, \lambda_x)\right)^2$, giving ML estimates $(\hat{\lambda}_y, \hat{\lambda}_x)$. We use the simpler $Y^{\lambda_y}$ and $X^{\lambda_x}$ (or $\log X$, $\log Y$). For MLR the process is the same, just more $\lambda$'s to estimate.

*Remark* (Box–Cox Likelihood — **not testable**, reference only). Issue: $\mathbf{Y} \mid \mathbf{X} \not\sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$; we need $\Psi_M$ so that $\Psi_M(\mathbf{Y} \mid \mathbf{X}) \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. Modify the log-likelihood and add the Jacobian determinant term (multivariate transformation method, STA237):

$$-\frac{n}{2}\ln(2\pi) - \frac{n}{2}\ln(\sigma^2) - \frac{1}{2\sigma^2}\sum_{i=1}^{n}\left(\Psi(y_i) - \beta_0 - \beta_1 x_i\right)^2 + n(\lambda - 1)\log\left(\mathrm{gm}(\mathbf{Y})\right)$$

and maximize over $\lambda$ (in R).

### Box–Cox in Practice

- The ML process for $\lambda$ has no closed form, so we use R.
- The actual Box–Cox transformations are hard to work with, so use **simpler powers**: much easier to **interpret** and apply, still close to normality, and you avoid figuring out what your variable now means.
- Pick something simpler than $\hat{\lambda}$, e.g. $\hat{\lambda} = 0.103 \Rightarrow$ use $\ln(Y)$ (close to 0):

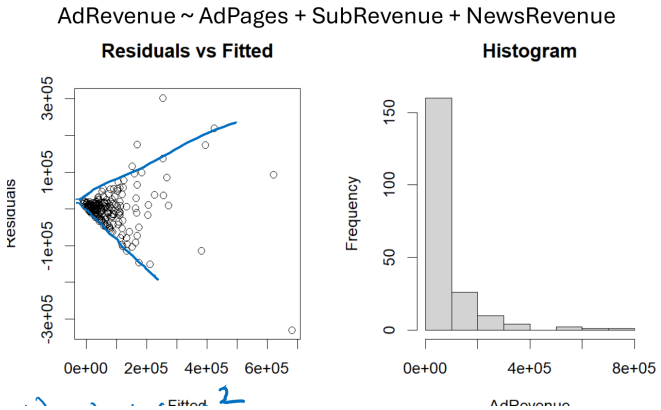
| $\lambda$ | transformation |
|---|---|
| 0.5 | square root |
| 0.33 | cube root |
| 0.25 | fourth root |
| 0 | log |
| $-0.5$ | reciprocal square root |
| $-1$ | reciprocal (inverse) |

- R reports confidence intervals for $\lambda$ — use them as a range of possibilities.

## Application and Interpretation of Transformed Models

**Example 29 Magazine Data**

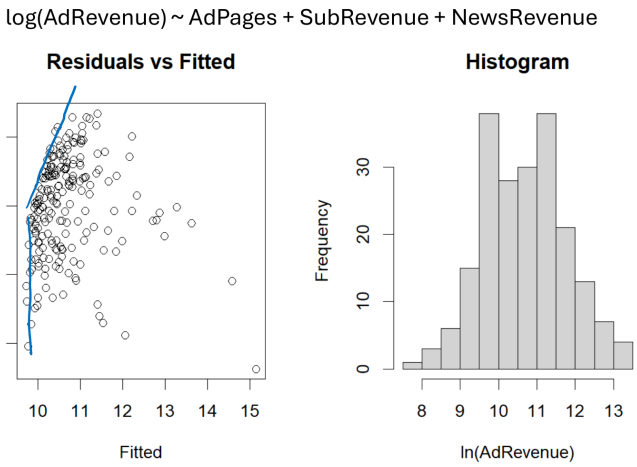
`mag <- read.csv("magazines.csv")`, 204 magazines. Predict $Y =$ `AdRevenue` from $X_1 =$ `AdPages`, $X_2 =$ `SubRevenue`, $X_3 =$ `NewsRevenue`.



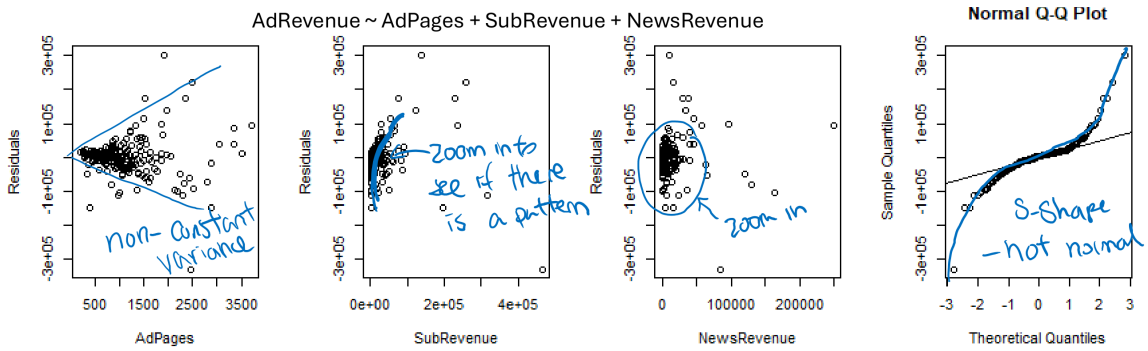

Issues: residuals fan out ($SD(Y) \propto E(Y)$ visually, so $\text{Var}(Y) \propto E(Y)^2$) and `AdRevenue` is strongly right-skewed $\Rightarrow$ **try a** $\log(Y)$ **transformation**.

**Example 30 Try** $\log($`AdRevenue`$)$

log and $\sqrt{}$ may improve right skew. Some improvement, but other violations may be present (the residuals still have a hard left edge).




More residual plots for the *original* model show many violations:

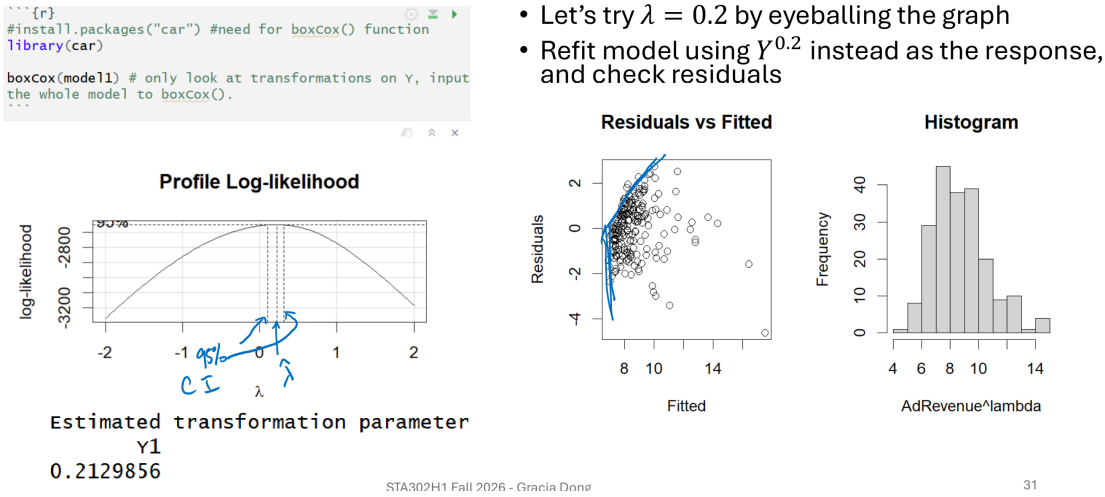



*Non-constant variance vs. AdPages; zoom in on SubRevenue/NewsRevenue clusters to see patterns; S-shaped QQ plot ⇒ not normal.* So let's try Box–Cox.

**Example 31 Box–Cox on $Y$**

Use `boxCox()` from the `car` package:

```
library(car)      # install.packages("car") for boxCox()
boxCox(model1)    # only transformations on Y; input the whole model
```




The profile log-likelihood peaks at $\hat{\lambda} = 0.213$ (dashed lines: $\hat{\lambda}$ and its 95% CI). Eyeballing, try $\lambda = 0.2$: refit with $Y^{0.2}$ as the response and check residuals. *Histogram is more symmetric, but the residuals vs. fitted still have an edge.*

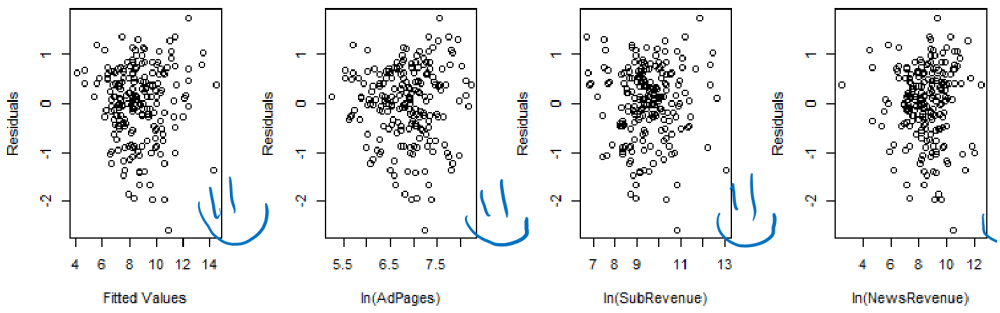
**Example 32 Box–Cox on $X$**

Use `powerTransform()` on the predictor columns:

```
p <- powerTransform(cbind(mag[,3:5]))  # all columns you may want to transform
summary(p)
```

| | Est Power | Rounded Pwr | Wald Lwr | Wald Upr |
|---|---|---|---|---|
| AdPages | 0.1119 | 0.00 | −0.0869 | 0.3107 |
| SubRevenue | −0.0084 | 0.00 | −0.0973 | 0.0804 |
| NewsRevenue | 0.0759 | 0.08 | 0.0106 | 0.1412 |

*Ignore the likelihood ratio tests in the output. I would personally round 0.08 again to 0.* Refit $Y^{0.2} \sim \log(X_1) + \log(X_2) + \log(X_3)$ ($Y^{0.2}$ from the previous slide) and check residuals:



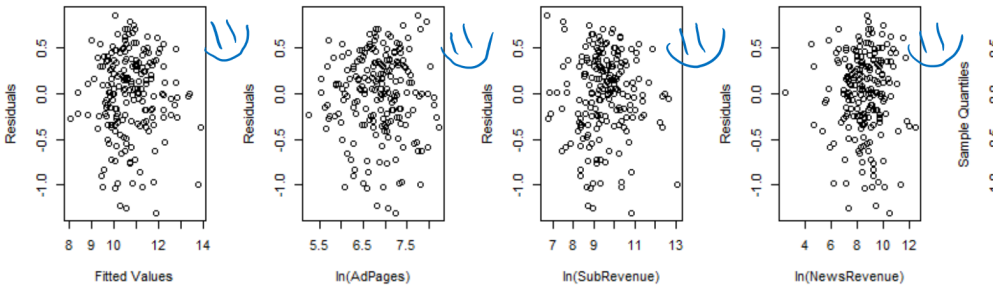

*Residual plots look fine; mild C shape in QQ plot — maybe OK, not super concerned.*

**Example 33 Box–Cox on $X$ and $Y$ Simultaneously**
Same function, now including the response column:

```
p <- powerTransform(cbind(mag[,2:5]))  # include X and Y here
summary(p)
```

| | | Est Power | Rounded Pwr | Wald Lwr | Wald Upr |
|---|---|---|---|---|---|
| $Y$ | AdRevenue | 0.1071 | 0.11 | 0.0299 | 0.1843 |
| $X_1$ | AdPages | 0.0883 | 0.00 | −0.0755 | 0.2521 |
| $X_2$ | SubRevenue | −0.0153 | 0.00 | −0.0862 | 0.0557 |
| $X_3$ | NewsRevenue | 0.0763 | 0.08 | 0.0115 | 0.1410 |

*Round 0.11 and 0.08 again to 0.* Refit $\log(Y) \sim \log(X_1) + \log(X_2) + \log(X_3)$:




*Residual plots look fine; mild C shape in QQ plot.*

*Remark* (Interpreting Transformed Models).
- Always interpret using the variables *as they are*: $\log(Y)$, $\log(X)$, etc. If a predictor is squared, say "for a one-unit increase in $x^2$".
- When the response is transformed, predicted values are conditional means on the **transformed scale**, not the original one. We don't "back-transform" to the original measurement scale.

**Example 34 Interpreting the Magazine Model**
For $\log(Y) \sim \log(X_1) + \log(X_2) + \log(X_3)$:

- **Intercept**: the expected **log of Revenue** due to Advertising when **log Ad Pages**, **log Subscription Revenue** and **log Newsstand Revenue** are 0.
- **Slope**: for a one unit increase in the **log of Advertising Pages**, the slope is the expected change in **log of Advertising Revenue** for a fixed **log Subscription Revenue** and **log Newsstand Revenue**.

## Impact of Violations on Sampling Distributions

*Remark* (Expected Value Vector and Covariance Matrix). For a random column vector $\mathbf{Y} = (Y_1, \ldots, Y_n)'$: $E[\mathbf{Y}] = (E[Y_1], \ldots, E[Y_n])' = \boldsymbol{\mu}$, and

$$\text{Var}(\mathbf{Y}) = \text{Cov}(\mathbf{Y}) = E\big[(\mathbf{Y} - E[\mathbf{Y}])(\mathbf{Y} - E[\mathbf{Y}])'\big]$$

is $n \times n$, with $(i,j)$-th entry $\text{Cov}(Y_i, Y_j)$ and $i$-th diagonal entry $\text{Var}(Y_i)$.

**Proposition 35 Linear Transformations.**
Let $\mathbf{Y}$ have mean $\boldsymbol{\mu}$ and covariance $\boldsymbol{\Sigma}$, and let $\mathbf{A}$ $(n \times p)$ and $\mathbf{a} \in \mathbb{R}^p$ be constants. Then

$$\begin{aligned} &\text{Var}(\mathbf{Y}) = E(\mathbf{Y}\mathbf{Y}') - \boldsymbol{\mu}\boldsymbol{\mu}' && \textit{cf. } V(Z) = E(Z^2) - E(Z)^2 \\ &E(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\mu} + \mathbf{a} && \textit{cf. } E(cZ+d) = cE(Z) + d \\ &\text{Var}(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A} && \textit{cf. } V(cZ+d) = c^2 V(Z) \end{aligned}$$

If $\mathbf{Y} \sim N(\boldsymbol{\mu}, \boldsymbol{\Sigma})$, then $\mathbf{A}'\mathbf{Y} \sim N(\mathbf{A}'\boldsymbol{\mu}, \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A})$.

*Remark* (MLR Setup). Assuming the assumptions hold, $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}$ with $\boldsymbol{\epsilon} \sim N(\mathbf{0}_n, \sigma^2\mathbf{I}_n)$, i.e. $E[\boldsymbol{\epsilon}] = \mathbf{0}$, $\text{Var}[\boldsymbol{\epsilon}] = \text{diag}(\sigma^2, \ldots, \sigma^2)$. So $\mathbf{Y}$ is multivariate normal too: $\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$.
$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$: $\mathbf{X}$ is fixed and $\mathbf{Y}$ random, so $\hat{\boldsymbol{\beta}}$ is random (a function of $\mathbf{Y}$); $\boldsymbol{\beta}$ is an unknown fixed parameter. Note $\hat{\boldsymbol{\beta}} = \mathbf{A}'\mathbf{Y}$ with $\mathbf{A}' = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

**Proposition 36 $\hat{\beta}$ is Unbiased.**

$$\begin{aligned} E(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) &= E((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'E(\mathbf{Y} \mid \mathbf{X}) \\ &= (\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})\boldsymbol{\beta} = \boldsymbol{\beta} \end{aligned}$$

**Proposition 37 Variance of $\hat{\beta}$.**
Using $\text{Var}(\mathbf{A}'\mathbf{Y}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A}$ with $\boldsymbol{\Sigma} = \text{Var}(\mathbf{Y} \mid \mathbf{X}) = \sigma^2\mathbf{I}$ and $\mathbf{A} = \left((\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\right)' = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}$ *($\mathbf{X}'\mathbf{X}$, hence its inverse, is symmetric)*:

$$\text{Var}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}(\mathbf{X}'\mathbf{X})(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2(\mathbf{X}'\mathbf{X})^{-1}$$

*$\sigma^2$ is a scalar, so it can be pulled out front.*

**Theorem 38 Sampling Distribution of $\hat{\beta}$.**
By linearity of normal random variables,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N_{p+1}(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1})}$$

So when the assumptions hold, the estimators are

- **unbiased** (on average estimate the value they should),
- generally **correlated** (off-diagonals of $(\mathbf{X}'\mathbf{X})^{-1}$ not guaranteed to be 0),
- dependent on the same constant variance $\sigma^2$ as the errors.

## Violations Break the Sampling Distribution

- **Linearity**: the mean of $\hat{\boldsymbol{\beta}}$ is no longer $\boldsymbol{\beta} \Rightarrow$ estimates are biased.
- **Constant variance**: no longer a single $\sigma^2$ in the variance.
- **Uncorrelated errors**: the variance of the estimators will be under- or over-estimated.
- **Normality**: estimators won't have a normal sampling distribution.

When assumptions are violated, we don't know the correct way to describe variation due to sampling $\Rightarrow$ **inference will be incorrect or misleading**.

## Gauss–Markov Theorem

**Theorem 39 Gauss–Markov (Gauss 1821, Markov 1900).**
Let $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$ *(linearity)*, $\mathbf{X}$ full rank, and $\mathrm{Var}(\mathbf{Y}) = \sigma^2\mathbf{I}$ *(constant variance, uncorrelated errors)*. Then the OLS estimator $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ is the **Best Linear Unbiased Estimator (BLUE)** of $\boldsymbol{\beta}$: for any $\mathbf{a}'\boldsymbol{\beta}$, among estimators that are linear in $\mathbf{Y}$ and unbiased, $\mathbf{a}'\hat{\boldsymbol{\beta}}$ has **minimum variance**. *OLS is the "best" — lowest sampling variance in the class of linear unbiased estimators.*

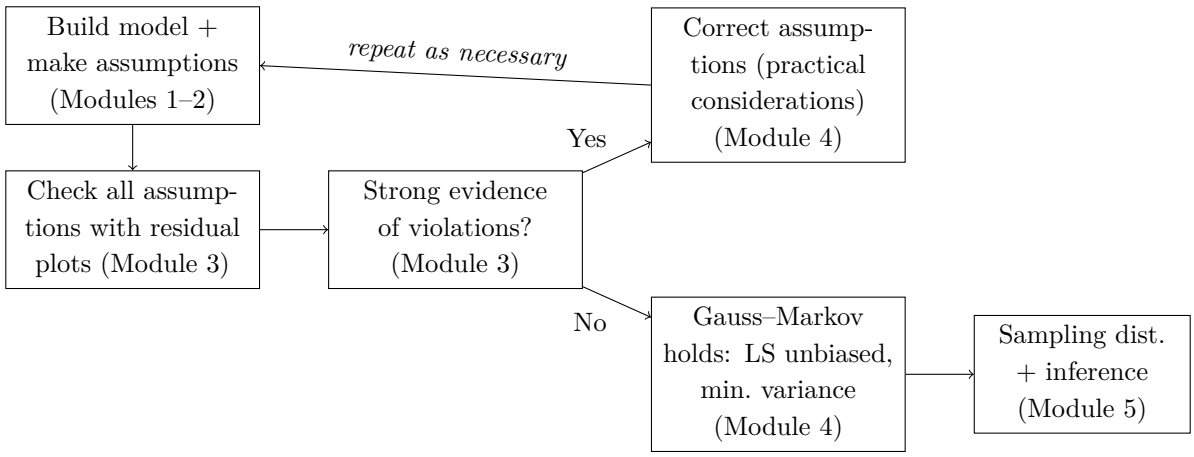
| **Ordinary Least Squares** | **Maximum Likelihood** |
|---|---|
| Minimizes $RSS = (\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}})'(\mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}) = \sum_{i=1}^{n} \hat{e}_i^2$ | Maximizes $\ln L(\boldsymbol{\beta}, \sigma^2 \mid \mathbf{Y}, \mathbf{X}) \propto -\frac{1}{2\sigma^2}\sum_{i=1}^{n}(y_i - \mathbf{x}_i\boldsymbol{\beta})^2$ *(the sum is the RSS)* |
| $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ | $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ |
| Requires: linearity, constant variance, uncorrelated errors. **No** normality assumption. | Requires: linearity, constant variance, uncorrelated errors, **and normality**. |

### Which Estimation Procedure to Use?

- If normality does not hold, OLS still gives an unbiased and most precise estimator of $\boldsymbol{\beta}$.
- If normality holds, MLE and OLS are equivalent (and both BLUE).
- OLS is **always** more precise than any other estimator linear in $\mathbf{Y}$.
- If inference (hypothesis tests, CIs) isn't important, OLS is the better choice — but the point of statistics is to do inference.

*Ideally, normality is satisfied so OLS and MLE are the same.*

## Summary




### Approaches to Transformations

- **Variance stabilizing transformation**

  - **Only** on $Y$; corrects non-constant variance (can improve other violations, but can also make them worse).
  - No obvious choice; common: $\ln(y-c)$, $\sqrt{y-c}$ (variance usually increases with $Y$).
  - Use judgement from the plots and the funnel direction (e.g. $\ln(c-y)$).
- **Box–Cox transformation** (on $Y$ and/or $X$)
  - Corrects non-normality in $Y$ and/or $X$ (sometimes helps non-linearity).
  - Estimate $Y^{\lambda}$ and/or $X^{\lambda}$ by maximum likelihood.
  - Guide for simpler transformations: $\hat{\lambda}=0 \to \ln(\cdot)$; $\hat{\lambda}\neq 0 \to Y^{\hat{\lambda}}$ or $X^{\hat{\lambda}}$.

  *Generally: transform $X$ for linearity issues.*
- We **cannot** correct correlated errors after data collection — it **must** be addressed at the study design / data collection stage.

**Workflow**

1. Plot data: scatterplot, scatterplot matrix (pairs plot), histogram, ...
2. Plot residuals: all residual plots, QQ plot, ...
3. Fix the **worst** violation first with an appropriate transformation.
4. Check residual plots again after *every* transformation.
5. Avoid complex/too many transformations — keep things interpretable, and keep in mind what variables represent.

*Weisberg Ch 8 has a good discussion on transformations.*

*Remark* (Module Takeaways).
- Why does a variance stabilizing transformation correct non-constant variance?
- What is a Box–Cox transformation trying to achieve?
- How do we select and implement transformations to correct model violations?
- What does this change in how we interpret model coefficients?
- Why is it important to attempt to correct model violations?
- Why is the OLS estimator the "best" estimator?

*Next week: inference on coefficients and predicted values. The first hour of Module 5 (inference on coefficients) is the last material on the term test.*

# Week 5: Inference on Linear Regression Components

*Remark* (Where We Are). Build model + make assumptions → check all assumptions with residual plots → strong evidence of violations? If **yes**, correct the assumptions and repeat. If **no**, Gauss–Markov holds (LS estimates are unbiased with minimum variance) ⇒ **today**: use this to build sampling distributions and make inference.

**Module Outline**

- Sampling distribution of coefficients
- Inference on coefficients: confidence intervals and hypothesis tests
- Inference for the mean response
- Predictions for actual responses

## Sampling Distribution of $\hat{\beta}$

*Remark* (Why Sampling Distributions).

- When we estimate anything, we need to quantify its error: how much will $\hat{\boldsymbol{\beta}}$ change from sample to sample?
- Defining a sampling distribution uses **assumptions about the population**. If the assumptions don't hold, the sampling distribution is wrong.
- The sampling distribution is our reference for what is considered normal variation to expect; we compare estimates to it to decide whether a value is reasonable.

**Definition - Pivotal Quantity**

Always of the form

$$\boxed{\frac{\text{Estimator} - \text{Truth}}{\text{Standard Error}}} \qquad \text{e.g.} \qquad \frac{\hat{\beta}_j - \beta_j}{SE(\hat{\beta}_j)}$$

It is the basis of confidence intervals and hypothesis tests. *E.g. the z-score of the sample mean,* $z = \frac{\bar{X}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$ *(STA238/220).*

**Theorem 40 Sampling Distribution of $\hat{\beta}$ (from Week 4).**
From the assumptions $\mathbf{Y} \mid \mathbf{X} \sim N(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$ and $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$, by properties of expectation, variance and linearity of the normal,

$$\boxed{\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta},\ \sigma^2(\mathbf{X}'\mathbf{X})^{-1})}$$

So when the assumptions hold, the estimators are

- **unbiased** (on average estimate the value they should),
- generally **correlated** (off-diagonals of $(\mathbf{X}'\mathbf{X})^{-1}$ not guaranteed to be 0),
- dependent on the same constant variance $\sigma^2$ as the errors.

**Definition - Covariance Matrix of $\hat{\beta}$**

For MLR:

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \begin{pmatrix} \text{Var}(\hat{\beta}_0 \mid \mathbf{X}) & \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) & \cdots & \text{Cov}(\hat{\beta}_0, \hat{\beta}_p \mid \mathbf{X}) \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_1 \mid \mathbf{X}) & \text{Var}(\hat{\beta}_1 \mid \mathbf{X}) & \cdots & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p \mid \mathbf{X}) \\ \vdots & \vdots & \ddots & \vdots \\ \text{Cov}(\hat{\beta}_0, \hat{\beta}_p \mid \mathbf{X}) & \text{Cov}(\hat{\beta}_1, \hat{\beta}_p \mid \mathbf{X}) & \cdots & \text{Var}(\hat{\beta}_p \mid \mathbf{X}) \end{pmatrix}$$

For SLR:

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \frac{\sigma^2}{\sum(x_i - \bar{x})^2} \begin{pmatrix} \sum x_i^2/n & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix}$$

The estimated coefficients have non-zero covariance.

*Remark* (Using the Covariance Matrix). Recall $\text{Var}(\mathbf{A}'\mathbf{Y} + \mathbf{a}) = \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A}$ if $\text{Var}(\mathbf{Y}) = \boldsymbol{\Sigma}$. We will often need the variance of expressions such as $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$ for inference, interpretation and prediction (later this module).

**Exercise 41 Variance of the Residuals.**
Show $\text{Var}(\hat{\epsilon}_i \mid \mathbf{X}) = \sigma^2(1-h_{ii})$ (Week 3), where $h_{ii}$ is the $i$-th diagonal element of $\mathbf{H} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.
*Hints: work with the vector* $\hat{\boldsymbol{\epsilon}} = \mathbf{Y} - \hat{\mathbf{Y}}$*; write* $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$*,* $\hat{\boldsymbol{\epsilon}} = (\mathbf{I} - \mathbf{H})\mathbf{Y}$*;* $\mathbf{H}^2 = \mathbf{H}$ *and* $(\mathbf{I} - \mathbf{H})^2 = ?$
$(\mathbf{I} - \mathbf{H})^2 = \mathbf{I} - 2\mathbf{H} + \mathbf{H}^2 = \mathbf{I} - \mathbf{H}$, and $\mathbf{I} - \mathbf{H}$ is symmetric. With $\mathbf{A}' = \mathbf{I} - \mathbf{H}$,

$$\text{Var}(\hat{\boldsymbol{\epsilon}} \mid \mathbf{X}) = (\mathbf{I} - \mathbf{H})\,\sigma^2\mathbf{I}\,(\mathbf{I} - \mathbf{H})' = \sigma^2(\mathbf{I} - \mathbf{H})^2 = \sigma^2(\mathbf{I} - \mathbf{H}),$$

whose $i$-th diagonal entry is $\sigma^2(1 - h_{ii})$.

**The $\hat{\beta}_j$'s Are Correlated**

- Generally $\text{Cov}(\hat{\beta}_i, \hat{\beta}_j \mid \mathbf{X}) \neq 0$ — look at the $(i, j)$-th element of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$.
- The more correlated the predictors $X_i$ and $X_j$ are in the sample, the more correlated **in the opposite direction** $\hat{\beta}_i$ and $\hat{\beta}_j$ are.
- We want to avoid **multicollinearity**: predictors that are [close to] linearly dependent. Perfect multicollinearity makes $\mathbf{X}'\mathbf{X}$ not invertible.

**Example 42 Correlated $\hat{\beta}$'s: House Prices**
True model: $Y = 1 + 3X_1 + 2X_2$ (in \$100,000), $X_1 = \#$ bedrooms, $X_2 = \#$ bathrooms. The dataset is almost exclusively condos with the same number of bedrooms and bathrooms: $X_1 \approx X_2$, so $\text{corr}(X_1, X_2) \approx 1$.

- $\hat{Y} = 1 + 3X_1 + 2X_2$, $\hat{Y} = 1 + 4X_1 + 1X_2$, $\hat{Y} = 1 + 5X_1 + 0X_2$, $\hat{Y} = 1 + 6X_1 - 1X_2$, … are all likely and valid on the training data $\Rightarrow \text{corr}(\hat{\beta}_1, \hat{\beta}_2) \approx -1$.
- The more positively correlated $X_1$, $X_2$ are, the more **negatively** correlated $\hat{\beta}_1$, $\hat{\beta}_2$ are, **and** the higher the variance of $\hat{\beta}_1$ and $\hat{\beta}_2$.

*Next week (feature selection): we probably don't need both* $X_1$ *and* $X_2$.

## Unknown $\sigma^2$ and the $t$ Distribution

**Definition - Estimating $\sigma^2$ (Week 3)**

$\sigma^2$ is an unknown population parameter; estimate it with

$$\boxed{s^2 = \hat{\sigma}^2 = \underbrace{\frac{1}{n-(p+1)}}_{1/df} \underbrace{\sum_{i=1}^{n} (Y_i - \hat{Y}_i)^2}_{RSS} = \frac{\hat{\boldsymbol{\epsilon}}'\hat{\boldsymbol{\epsilon}}}{n-(p+1)} = \frac{RSS}{df}}$$

where $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_{i1} + \cdots$. So we use $s^2(\mathbf{X}'\mathbf{X})^{-1}$ instead of $\sigma^2(\mathbf{X}'\mathbf{X})^{-1}$ as $\text{Var}(\hat{\boldsymbol{\beta}})$ — but this **adds additional sampling variation**. *Intercept-only model $y = \beta_0 + \epsilon$ gives $\hat{\beta}_0 = \bar{y}$: this is exactly inference on the mean from last year.*

*Remark* (Analogy with the Sample Mean). If $y_i \overset{iid}{\sim} N(\mu, \sigma^2)$ then $\bar{y} \sim N(\mu, \sigma^2/n)$, i.e. $\frac{\bar{y}-\mu}{\sigma/\sqrt{n}} \sim N(0,1)$. But in practice we used $\frac{\bar{y}-\mu}{s/\sqrt{n}} \sim T_{n-1}$. *Recall* $T = \dfrac{N(0,1)}{\sqrt{\chi^2_{df}/df}}$.

**Theorem 43 Pivotal Quantity for $\hat{\boldsymbol{\beta}}$.**
Since $\hat{\boldsymbol{\beta}} \sim N(\boldsymbol{\beta}, \sigma^2(\mathbf{X}'\mathbf{X})^{-1})$,

$$\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sqrt{\sigma^2(\mathbf{X}'\mathbf{X})^{-1}}} \sim N(0,1) \qquad \text{and} \qquad \boxed{\frac{\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}}{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}}} \sim T_{n-p-1}}$$

We use this to build confidence intervals and perform hypothesis tests. *Non-standard notation: the square root is* elementwise *(taken on the diagonal), since we think about each $\hat{\beta}_j$ separately.*

## Confidence Intervals and Hypothesis Tests for $\beta_j$

*Remark* (General Form). Many inferential procedures take a common form:

- **CIs**: **estimate** $\pm$ (critical value)$\times$(standard error)
- **Test statistics**: $\dfrac{\text{estimate} - \text{truth}}{\text{standard error}}$ *("truth" = the null value)*

The sampling distribution has all of these components:

$$\frac{\overbrace{\hat{\boldsymbol{\beta}}}^{\text{estimate}} - \boldsymbol{\beta}}{\underbrace{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}}}_{\text{standard error}}} \sim T_{n-p-1} \qquad \text{cf.} \quad \frac{\bar{y}-\mu}{s/\sqrt{n}} \sim T_{n-1}$$
$$\text{critical value} \uparrow$$

**Definition - $100(1-\alpha)\%$ Confidence Interval for $\beta_j$**

$$\hat{\beta}_j \pm t_{\alpha/2,\, n-p-1}\; s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}}$$

- $\alpha$ is the chosen significance level; $1-\alpha$ is the confidence level.
- The **critical value** is the $\alpha/2$ quantile of the $T$ distribution with $n-p-1$ degrees of freedom. *In R:* `qt(alpha/2, df)` *for the negative critical value,* `qt(1-alpha/2, df)` *for the positive one.*
- $(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}$ is the $(j+1)$-th element of the main diagonal, e.g. $\text{Var}(\hat{\beta}_0) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)}$.

*Compare* $\bar{y} \pm t_{\alpha/2,n-1}\, s/\sqrt{n}$.

### Confidence Interval Interpretation

- Built using the data and the sampling distribution.
- A window $2 \times t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimate $\hat{\beta}_j$. We would see the true $\beta_j$ in this window $(1-\alpha)\%$ of the time if repeated samples are taken.
- **Interpretation**: $(1-\alpha)\%$ of all intervals computed using data repeatedly obtained from the same population would contain the true $\beta_j$. *Can also say: plausible values of* $\beta_j$ *with* $(1-\alpha)\%$ *confidence.*

**Definition - Hypothesis Test of** $H_0 : \beta_j = \beta_j^0$

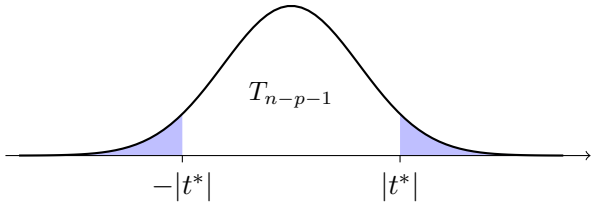
Test statistic:

$$t^* = \frac{\hat{\beta}_j - \beta_j^0}{s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}}} = \frac{\text{estimate} - \text{null value}}{SE(\text{estimate})} \qquad \text{cf.} \quad \frac{\bar{y}-\mu_0}{s/\sqrt{n}}$$

- $\beta_j^0$ is the **null value**: usually 0 for regression, but can be anything. Most common: $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$ — "do I need $X_j$ in the model?"
- The standard error $s\sqrt{(\mathbf{X}'\mathbf{X})^{-1}_{(j+1,j+1)}} = SE(\hat{\beta}_j)$ standardizes the difference between the estimate and the hypothesized value.
- Compare to the expected variability under $T_{n-p-1}$ to see whether the hypothesized value is plausible.

### Hypothesis Test Interpretation

- $H_0 : \beta_j = 0$ tests the null that **no linear relationship exists between $X_j$ and $Y$** (in the presence of the other predictors).
- Under $H_0$, $t^* \sim T_{n-p-1}$. Conclude by comparing to the sampling distribution:
  - If $|t^*| > t_{\alpha/2,n-p-1}$, reject $H_0$.
  - If $P(|T_{n-p-1}| \geq |t^*|) < \alpha$, reject $H_0$. *In R:* `2*pt(-abs(tstar), df)` *(both tails).*
- If rejected: claim a significant linear relationship exists between $X_j$ and $Y$ in the presence of the other predictors.




*Two-sided p-value: shaded area in both tails.*

*Remark* (CIs and Tests Agree). The usual relationship applies: if a 95% CI does not contain the hypothesized value, the null hypothesis would be rejected at the 0.05 significance level.

## Example: Room Cleaning Dataset

**Example 44 Setup**
Data from Foster, Stine, and Waterman (1997, p. 9): 53 observations. Model the number of rooms cleaned ($Y$) by the size of the cleaning crew ($X$). *From the scatterplot, constant variance is mildly violated (spread grows with crew size).*
Step 0: read in data, build the design matrix and response vector (so we can also compute manually):

```
data <- read.table("cleaning.txt", header=T)
Y <- data[,3]                                  # Rooms Cleaned
X <- as.matrix(cbind(rep(1, 53), data[,2]))    # Predictor: crew size
```

**Example 45 Built-in R:** `summary()`
```
model <- lm(Rooms ~ Crews, data=data)
summary(model)

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   1.7847     2.0965   0.851    0.399
Crews         3.7009     0.2118  17.472   <2e-16 ***
Residual standard error: 7.336 on 51 degrees of freedom
Multiple R-squared: 0.8569, Adjusted R-squared: 0.854
F-statistic: 305.3 on 1 and 51 DF, p-value: < 2.2e-16
```

- `Estimate`: $\hat{\beta}_j$; `Std. Error`: $SE(\hat{\beta}_j) = \sqrt{\text{Var}(\hat{\beta}_j)}$
- `t value`: test statistic $t^* = \hat{\beta}_j / SE(\hat{\beta}_j)$
- `Pr(>|t|)`: p-value for $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$
- `Residual standard error`: $s$; degrees of freedom $= n - p - 1 = 53 - 2 = 51$

**Example 46 CIs from the Summary Output**
CIs are always **estimate** $\pm$ (critical value)$\times$(standard error), with estimate $= \hat{\beta}_j$, critical value = `qt(alpha/2, df)` or `qt(1-alpha/2, df)`, SE $= SE(\hat{\beta}_j)$:

```
summ  <- summary(model)
```

```
coefs <- summ$coefficients   # the 4 columns of summary output
U <- coefs[,1] + qt(0.975, df = 51)*coefs[,2]   # upper bound
L <- coefs[,1] - qt(0.975, df = 51)*coefs[,2]   # lower bound
cbind(L, U)
#                    L        U
# (Intercept) -2.424160 5.993558
# Crews        3.275653 4.126134
```

**Example 47 CIs with `confint()`**
Defaults to 95% CIs; works with many model objects (GLMs, nonlinear/weighted least squares, ...); can specify which parameters (default: all).

```
confint(model)                  # 95%
confint(model, level=0.9)       # 90%: Crews [3.346039, 4.055748]
confint(model, parm = "Crews")  # only beta1
confint(model, parm = 1)        # by order; 1 is the intercept
```

*Remark* (Term Test Coverage). **Term test coverage ends here.**

### Study Tips: Transformations and Assumption Checking

- Practice with different datasets.
- Generate datasets from distributions that violate the assumptions and check residual plots to build intuition.
- Try different transformations: which violations get fixed? Which get worse?
- Generate datasets that *satisfy* the assumptions and look at their residual plots — what does an acceptable amount of noise/deviation look like?

### Rough Guidelines for Transformations

- No universal checklist; it varies from problem to problem.
- Issues with **linearity**: consider transforming the **predictors**.
- Issues with **residual variance/distribution**: consider transforming the **response** to stabilize the variance and/or fix the residual distribution.

**Example 48 Room Cleaning: Manual Calculations**
**Step 1**: $\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$ gives the fitted model $\hat{y}_i = 1.785 + 3.701x_i$.

```
xtx_inv  <- solve(t(X) %*% X)   # (X'X)^-1
xty      <- t(X) %*% Y          # X'Y
beta_hat <- xtx_inv %*% xty     # 1.784699, 3.700893
```

**Step 2**: RSS and $s^2$:

$$RSS = 2744.796, \qquad s^2 = \frac{RSS}{n-p-1} = \frac{2744.796}{53-1-1} = 53.82$$

```
Y_hat <- X %*% beta_hat
RSS   <- sum((Y_hat - Y)^2)      # 2744.796
RSS/51                           # 53.81953
```

**Step 3**: variances. `xtx_inv` $= \begin{pmatrix} 0.081666 & -0.007235 \\ -0.007235 & 0.000834 \end{pmatrix}$, so

$$\text{Var}(\hat{\beta}_0) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)} = 53.82(0.08167) = 4.395$$
$$\text{Var}(\hat{\beta}_1) = s^2(\mathbf{X}'\mathbf{X})^{-1}_{(2,2)} = 53.82(0.00083) = 0.0449$$

*`s2*xtx_inv` also gives* $\text{Cov}(\hat{\beta}_0, \hat{\beta}_1) = -0.3894$.

**Step 4**: critical value and distribution. Sampling distribution is $T_{n-p-1} = T_{51}$, $\alpha = 0.05$: $|t_{0.025,51}| = 2.00$ *(`qt(0.025, 51)` $= -2.007584$)*.

**Step 5**: confidence interval:

$$\hat{\beta}_0 \pm t_{0.025,51}\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}_{(1,1)}} = 1.785 \pm 2.00\sqrt{4.395} = [-2.42,\ 5.99]$$

**Interpretation**: we are 95% confident that $\beta_0$ is between $-2.42$ and 5.99. *If we repeat this process with repeated samples, 95% of the created CIs will contain the true $\beta_0$.*

```
beta_hat[1] + crit_val*sqrt((s2*xtx_inv)[1,1])   # -2.42416
beta_hat[1] - crit_val*sqrt((s2*xtx_inv)[1,1])   #  5.993558
```

**Step 5 (test)**: in linear regression we almost always test $H_0 : \beta_j = 0$ vs. $H_A : \beta_j \neq 0$.

$$H_0 : \beta_1 = 0 : \qquad t^* = \frac{\hat{\beta}_1 - 0}{\sqrt{s^2(\mathbf{X}'\mathbf{X})^{-1}_{(2,2)}}} = \frac{3.701}{\sqrt{0.0449}} = 17.47$$

Either compute the p-value or compare with the critical value: $17.47 \gg 2.00 \Rightarrow$ **reject** $H_0$; a statistically significant linear relationship exists.

```
beta_hat[2]/sqrt((s2*xtx_inv)[2,2])   # 17.47211
2*pt(-17.47, 51)                      # 3.573044e-23
```

**Exercise 49 Repeat for $\beta_1$ (CI) and $\beta_0$ (Test).**
CI for $\beta_1$: $3.701 \pm 2.0076\sqrt{0.04487} = 3.701 \pm 2.0076(0.2118) =$ **[3.276, 4.126]**, matching `confint()`.
Test for $\beta_0$: $t^* = 1.785/\sqrt{4.395} = 1.785/2.0965 = 0.851 < 2.00$, p-value $= 0.399 > 0.05 \Rightarrow$ **fail to reject** $H_0 : \beta_0 = 0$ (consistent with the CI $[-2.42, 5.99]$ containing 0).

**Exercise 50 Without Matrices.**
Using the data file, repeat the example with the SLR formulas, recalling

$$\text{Cov}(\hat{\boldsymbol{\beta}} \mid \mathbf{X}) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}\begin{pmatrix} \sum x_i^2/n & -\bar{x} \\ -\bar{x} & 1 \end{pmatrix},$$

i.e. $\text{Var}(\hat{\beta}_1) = \dfrac{s^2}{SS_{xx}}$ and $\text{Var}(\hat{\beta}_0) = s^2\dfrac{\sum x_i^2}{n\,SS_{xx}} = s^2\Big(\dfrac{1}{n} + \dfrac{\bar{x}^2}{SS_{xx}}\Big)$.

## Inference for the Mean Response $E(\mathbf{Y} \mid \mathbf{X})$

### Three Kinds of Inference with Linear Regression

- On $\boldsymbol{\beta}$: the relationship between the response and predictors *($\boldsymbol{\beta}$ are unknown population parameters)*.
- On $E(\mathbf{Y} \mid \mathbf{X})$, estimated by $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$ *(an unknown population parameter, similar to inference on $\mu$ previously)*.
- **Prediction** about an individual observation's $y_0$ *($y_0$ is an unknown future value)*.

**Definition - Estimated Mean Response**

For each $\mathbf{x}_0' = (1, x_1, x_2, \ldots, x_p)$, the estimate is

$$\hat{y}_0 = \hat{E}(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0') = \mathbf{x}_0'\hat{\boldsymbol{\beta}} = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{Y}$$

This is a **linear combination of $\mathbf{Y}$**, so we can use $\mathbf{Y} \mid \mathbf{X} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I}_n)$ and $\mathbf{A}'\mathbf{Y} \sim N(\mathbf{A}'\boldsymbol{\mu}, \mathbf{A}'\boldsymbol{\Sigma}\mathbf{A})$ with $\mathbf{A}' = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'$.

**Proposition 51 Sampling Distribution of $\hat{y}_0$.**

Mean:

$$E(\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0) = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\mathbf{X}\boldsymbol{\beta} = \mathbf{x}_0'\boldsymbol{\beta}$$

Variance, with $\mathbf{A} = \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$:

$$\mathrm{Var}(\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0) = \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\,\sigma^2\mathbf{I}\,\mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

*Equivalently* $\mathrm{Var}(\mathbf{x}_0'\hat{\boldsymbol{\beta}}) = \mathbf{x}_0'\,\mathrm{Var}(\hat{\boldsymbol{\beta}})\mathbf{x}_0$. Hence

$$\boxed{\hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0 \sim N\big(\mathbf{x}_0'\boldsymbol{\beta},\ \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0\big)}$$

Same issue as before: $\sigma^2$ is unknown, so estimate with $s^2$, which adds sampling variability $\Rightarrow T$ distribution:

$$\boxed{\frac{\hat{y}_0 - \mathbf{x}_0'\boldsymbol{\beta}}{\sqrt{s^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim T_{n-p-1}}$$

**Definition - $(1-\alpha)\%$ CI for the Mean Response $\mathbf{x}_0'\boldsymbol{\beta}$**

$$\boxed{\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-p-1}\ s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}}$$

Same $T_{n-p-1}$ distribution as before; a window $2\,t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimate $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

For SLR the standard error simplifies to

$$\sqrt{\mathrm{Var}(\hat{y}_0 \mid X, x_0)} = s\sqrt{\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum(x_i - \bar{x})^2}}$$

**Exercise 52 SLR Standard Error from the Matrix Formula.**
With $\mathbf{x}_0 = (1, x_0)'$ and $SS_{xx} = \sum(x_i - \bar{x})^2$,

$$\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = \frac{1}{SS_{xx}}\Big(\frac{\sum x_i^2}{n} - 2x_0\bar{x} + x_0^2\Big) = \frac{1}{SS_{xx}}\Big(\underbrace{\frac{\sum x_i^2}{n} - \bar{x}^2}_{SS_{xx}/n} + (x_0 - \bar{x})^2\Big)$$

$$= \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{SS_{xx}}$$

*The CI is narrowest at $x_0 = \bar{x}$ and widens as $x_0$ moves away from it.*

**Definition - Hypothesis Test for the Mean Response**

For $H_0 : y_0 = y_0^0$,

$$t^* = \frac{\hat{y}_0 - y_0^0}{s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}} \sim T_{n-p-1} \text{ under } H_0$$

Think of the mean response as a parameter similar to $\mu$ (the conditional population mean for individuals with attributes $\mathbf{x}_0$). Reject $H_0$ if $|t^*| > t_{\alpha/2,n-p-1}$, or if $P(|T_{n-p-1}| \geq |t^*|) < \alpha$. *Not really done in practice — usually we create confidence intervals.*

**Example 53 Room Cleaning: CI for Mean Response**
*What is the average number of rooms cleaned with a cleaning crew of size 5?* Fitted model $\hat{y}_i = 1.785 + 3.701x_i$.

```
new <- data.frame(Crews=5)   # name must match the original data
predict(model, newdata=new, interval="confidence", level=0.95)
#        fit      lwr      upr
# 1 20.28917 17.73171 22.84662

new <- data.frame(Crews=c(4,2,5))   # several points at once
predict(model, newdata=new, interval="confidence", level=0.95)
#         fit       lwr      upr
# 1 16.588272 13.750660 19.42588
# 2  9.186485  5.699374 12.67360
# 3 20.289166 17.731710 22.84662
```

**Exercise 54 Verify the CI Manually.**
Use R to find $(\mathbf{X}'\mathbf{X})^{-1}$, $|t_{0.025,51}|$, $SS_{xx}$, $SS_{yy}$, $SP_{xy}$, $\bar{X}$, $\bar{Y}$, then complete the calculation with both the SLR and MLR formulas.
MLR: with $\mathbf{x}_0 = (1, 5)'$, $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.081666 - 2(5)(0.007235) + 25(0.000834) = 0.03015$, so

$$SE = \sqrt{53.82(0.03015)} = \sqrt{1.623} = 1.274, \qquad 20.289 \pm 2.0076(1.274) = \mathbf{[17.73,\ 22.85]}.$$

### Prediction Interval for an Actual Response $y_0$

*Remark* (Actual vs. Mean Response).

- The regression model gives predicted values **only** for $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ — estimates of conditional mean responses. We cannot exactly predict any individual response. $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ is a fixed but unknown population parameter.
- But an important goal of regression is **prediction**: for an individual with known $\mathbf{x}_0$, what is their $y_0$?
- The actual response $y_0$ is a **realization of a random variable** $Y \mid \mathbf{X} = \mathbf{x}_0$: it can take any possible value, and is typically not $E(\mathbf{Y} \mid \mathbf{X} = \mathbf{x}_0)$ ($y_0$ is not necessarily on the line/plane of best fit). So we must account for the random error as well.

**Definition - Prediction Error**

Using $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$ as the predicted value,

$$y_0 - \hat{y}_0 = \mathbf{x}_0'\boldsymbol{\beta} + \varepsilon_0 - \hat{y}_0 = \underbrace{(\mathbf{x}_0'\boldsymbol{\beta} - \hat{y}_0)}_{\text{fit of mean}} + \underbrace{\varepsilon_0}_{\text{random error}}$$

Two sources of error:

- the random error in the population, $\varepsilon_0$;
- how well the model fits the mean response, $\mathbf{x}_0'\boldsymbol{\beta} - \hat{y}_0 = \mathbf{x}_0'\boldsymbol{\beta} - \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.

**Proposition 55 Distribution of $y_0 - \hat{y}_0$.**
Assuming the model assumptions hold, $\varepsilon_0 \sim N(0, \sigma^2)$ for the new observation is independent of the sample $\mathbf{Y}$, hence of $\hat{y}_0$. Mean:

$$E(y_0 - \hat{y}_0) = \mathbf{x}_0'\boldsymbol{\beta} + 0 - \mathbf{x}_0'\boldsymbol{\beta} = 0$$

Variance (no covariance term, by independence):

$$\mathrm{Var}(y_0 - \hat{y}_0) = \mathrm{Var}(\varepsilon_0) + \mathrm{Var}(\hat{y}_0) = \sigma^2 + \sigma^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0$$

A linear combination of normals is normal, so

$$y_0 - \hat{y}_0 \mid \mathbf{X}, \mathbf{x}_0 \sim N\big(0,\ \sigma^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]\big)$$

Estimating $\sigma^2$ with $s^2$ adds sampling variability $\Rightarrow T_{n-p-1}$:

$$\frac{y_0 - \hat{y}_0}{\sqrt{s^2[1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0]}} \sim T_{n-p-1}$$

Notice the extra "+1" term in the variance (from $\varepsilon_0$).

**Definition - $(1-\alpha)\%$ Prediction Interval for $y_0$**

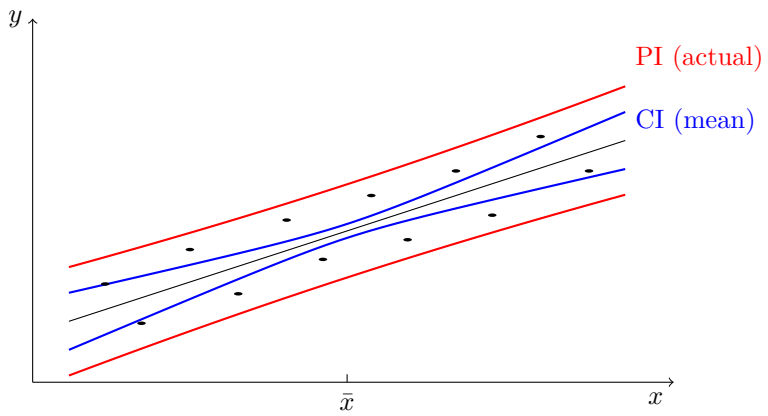
$$\mathbf{x}_0'\hat{\boldsymbol{\beta}} \pm t_{\alpha/2,\,n-p-1}\; s\sqrt{1+\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$$

- We cannot predict the exact value of $y_0$ since we do not know $\epsilon_0$; instead give a range of possible values for the actual response.
- **Not** a confidence interval: we are not estimating an unknown parameter — $y_0$ is a realization of a random variable.
- Same $T_{n-p-1}$ distribution; window $2\,t_{\alpha/2,n-p-1}$ standard errors wide, centred on the estimated mean response $\hat{y}_0 = \mathbf{x}_0'\hat{\boldsymbol{\beta}}$.
- **Wider** than the CI for the mean response $E(\mathbf{Y} \mid \mathbf{X}, \mathbf{x}_0)$.

For SLR:

$$\hat{\beta}_0 + \hat{\beta}_1 x_0 \pm t_{\alpha/2,\,n-2}\; s\sqrt{1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}}$$

*Follows from the previous exercise by adding 1 inside the root.*




*Both bands are centred on the fitted line and narrowest at $\bar{x}$; the PI is always wider.*

**Example 56 Room Cleaning: PI for Actual Response**

*How many rooms will this cleaning crew of size 5 clean?*

```
new <- data.frame(Crews=5)
predict(model, newdata=new, interval="prediction", level=0.95)
#        fit      lwr      upr
# 1 20.28917 5.340774 35.23756

new <- data.frame(Crews=c(4,2,5))
predict(model, newdata=new, interval="prediction", level=0.95)
#         fit       lwr      upr
# 1 16.588272  1.589410 31.58713
```

```
# 2  9.186485 -5.948698 24.32167
# 3 20.289166  5.340774 35.23756
```

The prediction intervals are **wider** than the confidence intervals, but the **centres are the same**.

**Exercise 57 Verify the PI Manually.**
Use R to find $(\mathbf{X}'\mathbf{X})^{-1}$, $|t_{0.025,51}|$, $SS_{xx}$, $SS_{yy}$, $SP_{xy}$, $\bar{X}$, $\bar{Y}$; do the rest by hand with both the SLR and MLR formulas.
Reusing $\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 0.03015$ at $x_0 = 5$:

$$SE = \sqrt{53.82(1 + 0.03015)} = \sqrt{55.44} = 7.446, \qquad 20.289 \pm 2.0076(7.446) = \mathbf{[5.34,\ 35.24]}.$$

## Confidence vs. Prediction Intervals

| **Confidence interval** | **Prediction interval** |
|---|---|
| For the **mean response** | For a **future/specific observation** |
| "What would be the average selling price for a house with characteristics $\mathbf{x}_0$?" | "What is the predicted sale price for *this* house with characteristics $\mathbf{x}_0$?" |
| SE: $s\sqrt{\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$ | SE: $s\sqrt{1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$ |

- Prediction intervals are **always wider** than confidence intervals *(think: variability of $y_i$ vs. $\bar{y}$)* — they include variation in estimating the conditional mean *and* variation around the conditional mean.
- Both are centred at $\mathbf{x}_0'\hat{\boldsymbol{\beta}}$.
- Some sources call both "confidence intervals" but specify whether the interest is a single future response or the mean response.

*Remark* (Module Takeaways).
- How did we determine the properties of the sampling distribution, and where did the assumptions play a role?
- Why do we use a $T$ distribution with the sampling distribution in practice?
- How do we compute confidence/prediction intervals and conduct hypothesis tests on regression components?
- How are the inferential procedures concluded?
- What is the difference between estimating a mean response and predicting an actual response?

*Next week: variance decomposition — how much of the total variation is explained by the predictors?*

# Week 6: Variance Decomposition Part 1

## Module Outline

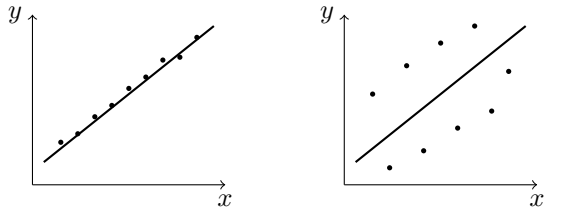
- Decomposition of the sum of squares
- ANOVA test for overall significance
- Partial $F$ test

Today we think about the **marginal** variance of $Y$, $\text{Var}(Y)$, in addition to the **conditional** variance of $Y \mid X$, $\text{Var}(Y \mid X)$.

## Variation in Linear Regression

*Remark* (Variation and Strength of a Linear Relationship).

- So far we treat the predictors $X$ as fixed and fit the model that minimizes the RSS (variation around the line of best fit).
- Variation tells us how strong the linear relationship is: **less noise around the trend** $\rightarrow$ stronger linear relationship $\rightarrow$ the predictor(s) explain the spread of $Y$ better.
- The $t$-test for $\beta_j$ also uses this variation around the line to decide whether there is a significant linear relationship between $X_j$ and $Y$ in the presence of the other predictors.




strong linear relationship  weak linear relationship

*Remark* (Variation in SLR and $t$-Tests). In SLR, $y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$.

- If $\beta_1 = 0$, then $y_i = \beta_0 + \varepsilon_i$ and $\hat{\beta}_0 = \bar{y}$: variation around a **horizontal line**. *$H_0 : \beta_1 = 0$ is the model $\hat{y}_i = \bar{y}$.*
- When there is a linear relationship, generally $\hat{y}_i \neq \bar{y}$.
- Testing $H_0 : \beta_1 = 0$ compares $\hat{\beta}_1 - 0$ to the variation in estimating $\hat{\beta}_1$: $\text{Var}(\hat{\beta}_1) = \frac{\sigma^2}{\sum(x_i - \bar{x})^2}$ is estimated by $\frac{s^2}{\sum(x_i - \bar{x})^2}$, which includes $s^2 = \frac{\sum(y_i - \hat{y}_i)^2}{n-2}$.
- For the $t$-test to indicate a significant linear relationship, the $\boldsymbol{\hat{y}_i - \bar{y}}$ (fitted line vs. horizontal line) across all observations would need to overall be larger than the residual differences $\boldsymbol{y_i - \hat{y}_i}$.

## Sum of Squares Decomposition

**Theorem 58 Law of Total Variance (STA237/220).**

$$V[Y] = \underbrace{E[V[Y \mid X]]}_{\text{unexplained}} + \underbrace{V[E[Y \mid X]]}_{\text{explained}}$$

If $X$ is discrete, the variance of $Y$ is the sum of the **within-group** (unexplained) variance and the **between-group** (explained by group membership) variance. In regression: the total variability of $Y$ is the variance that cannot be explained by $X$ plus the variance that can be explained by $X$. *Don't worry if you don't remember this.*

**Definition - Total Sum of Squares**

We are interested in explaining $\text{Var}(Y)$ *(not conditional on $X$)*: the total amount of variation in $Y$, quantified by

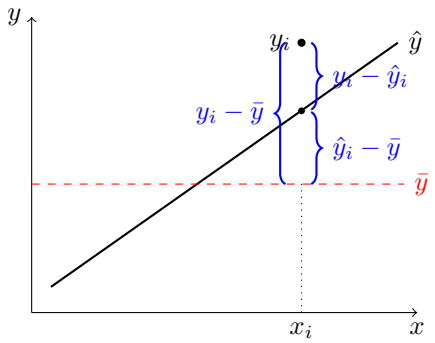
$$SST = SS_{yy} = \sum (y_i - \bar{y})^2,$$

the numerator of $\widehat{\text{Var}}(Y)$.

- Each predictor added to the model explains a little bit of this variation; how much depends on the slope $\hat{\beta}_j$.
- There is always some variation left over: the random error term can never be modelled.

Each deviation splits as

$$y_i - \bar{y} = \underbrace{(y_i - \hat{y}_i)}_{\text{residual}} + \underbrace{(\hat{y}_i - \bar{y})}_{\text{fitted vs. mean}}$$




*Graphical depiction of $y_i - \bar{y} = (y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})$ (after Sheather, p. 29).*

**Definition - The Three Sums of Squares**

- **Total** variation in $Y$ prior to fitting a model (ignoring predictors):

$$SST = \sum (y_i - \bar{y})^2 \ \propto \ \text{Var}(Y)$$

- **Explained** variation, by the predictors/the model:

$$SS_{reg} = \sum(\hat{y}_i - \bar{y})^2 \ \propto \ V(E(Y \mid X))$$

- **Unexplained** variation left over after fitting the model:

$$RSS = \sum(y_i - \hat{y}_i)^2 = \sum \hat{\epsilon}_i^2 \ \propto \ E(V(Y \mid X))$$

**Sum of squares decomposition** (analogous to the law of total variance):

$$\boxed{SST = SS_{reg} + RSS}$$

*Other sources may use different names and abbreviations for these.*

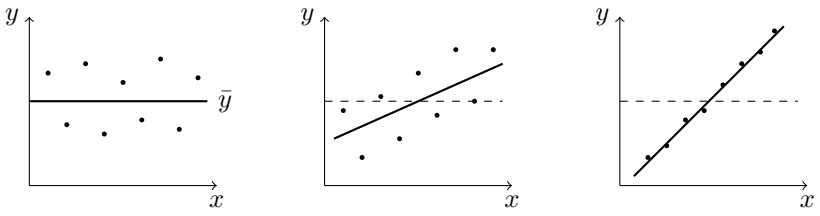
**Definition - Coefficient of Determination $R^2$**

The proportion of the variation in the response that is predictable from/explained by the predictor(s):

$$\boxed{R^2 = 1 - \frac{RSS}{SST} = \frac{SS_{reg}}{SST}}$$

In SLR, $R^2 = \text{corr}(X, Y)^2$.

*Remark* (Interpreting $SS_{reg}$).
- $SS_{reg}$ is how much of the overall variation in $Y$ is explained by $X$. If $X$ is categorical it is often called the **between-group** SS, and RSS the **within-group** SS.
- Sums of squares measure squared distance, like the numerator of a sample variance.
- If each $\hat{y}_i$ is generally **close** to $\bar{y}$, $SS_{reg} \approx 0$: the model isn't doing much to explain $Y$.
- If each $\hat{y}_i$ is generally **far** from $\bar{y}$, $SS_{reg}$ is large: the predictors explain a lot of what happens with $Y$.




$SS_{reg} \approx 0$
$RSS \approx SST$

$SS_{reg}$ small-ish

$SS_{reg}$ large
$RSS$ small

**Example 59 Adding a Predictor**
Data with a curved (decreasing, convex) trend:

- `y ~ x`: some points are far from the line of best fit, so RSS is somewhat large and $SS_{reg}$ somewhat large.

- `y ~ x + x^2`, $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x + \hat{\beta}_2 x^2$: **RSS decreases,** $SS_{reg}$ **increases**, and **SST stays the same** (the data did not change).

*Remark* (Picture: Conditional Distributions (Wikipedia ANOVA figures)). Picture the marginal distribution of $Y$ (spread $V(Y)$, centred at $E(Y) = \bar{y}$) next to the conditional distributions $Y \mid X = x_i$ (spreads $V(Y \mid X_i)$, centred at $E(Y \mid X = x_i)$).

- **No fit**: all $E(Y \mid X = x_i)$ sit at $\bar{y} \Rightarrow$ RSS big, $SS_{reg}$ small; $X$ does not explain how $Y$ changes.
- **OK fit**: the conditional means spread out $\Rightarrow$ $SS_{reg}$ bigger, RSS smaller.
- **Very good fit**: conditional means spread widely and each $V(Y \mid X_i)$ is narrow $\Rightarrow$ $SS_{reg}$ very large; $X$ explains a large part of the variance of $Y$.

## Proof of the Decomposition

*Remark* (Strategy). Show $\sum(y_i - \bar{y})^2 = \sum(\hat{y}_i - \bar{y})^2 + \sum(y_i - \hat{y}_i)^2$ by expanding the squares and showing the cross terms are 0. *Proofs of this type appear in many upper-year STA courses — decomposing sums of squares is a VERY important skill.*
Recall from Module 1: $\sum_{i=1}^n x_i \hat{\varepsilon}_i = 0$ and $\sum_{i=1}^n \hat{\varepsilon}_i = 0$. This holds for MLR too: the residuals are uncorrelated with every predictor and still sum to 0 (given an intercept). In matrix form, $\hat{\boldsymbol{\epsilon}}'\mathbf{X} = \mathbf{0}$.

**Proposition 60** $\hat{\boldsymbol{\epsilon}}'\mathbf{X} = \mathbf{0}$.
Using $\hat{\mathbf{Y}} = \mathbf{HY}$ and $\mathbf{H}' = \mathbf{H}$:

$$\begin{aligned}(\mathbf{Y} - \hat{\mathbf{Y}})'\mathbf{X} &= (\mathbf{Y}' - \hat{\mathbf{Y}}')\mathbf{X} = (\mathbf{Y}' - (\mathbf{HY})')\mathbf{X} = (\mathbf{Y}' - \mathbf{Y}'\mathbf{H}')\mathbf{X} \\ &= \mathbf{Y}'(\mathbf{I} - \mathbf{H})\mathbf{X} = \mathbf{Y}'(\mathbf{I} - \mathbf{X}(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}')\mathbf{X} \\ &= \mathbf{Y}'(\mathbf{X} - \mathbf{X}\underbrace{(\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{X}}_{\mathbf{I}}) = \mathbf{Y}'(\mathbf{X} - \mathbf{X}) = \mathbf{0}\end{aligned}$$

So $\sum_{i=1}^n x_{ij}\hat{\varepsilon}_i = 0$ for all $j$, and $\sum_{i=1}^n \hat{\varepsilon}_i = 0$ since the first column of the design matrix is $(1, \ldots, 1)'$.

*Proof of $SST = SS_{reg} + RSS$ (scalars).*

$$\sum(y_i - \bar{y})^2 = \sum\left((y_i - \hat{y}_i) + (\hat{y}_i - \bar{y})\right)^2 = \underbrace{\sum(y_i - \hat{y}_i)^2}_{RSS} + \underbrace{\sum(\hat{y}_i - \bar{y})^2}_{SS_{reg}} + \underbrace{2\sum(y_i - \hat{y}_i)(\hat{y}_i - \bar{y})}_{\text{show } =0}$$

With $y_i - \hat{y}_i = \hat{\varepsilon}_i$ and $\bar{y}$ not changing with $i$:

$$2\sum \hat{\varepsilon}_i(\hat{y}_i - \bar{y}) = 2\Big(\sum \hat{\varepsilon}_i\hat{y}_i - \bar{y}\underbrace{\sum \hat{\varepsilon}_i}_{0}\Big)$$

and

$$\sum \hat{\varepsilon}_i\hat{y}_i = \sum \hat{\varepsilon}_i(\hat{\beta}_0 + \hat{\beta}_1 x_{i1} + \hat{\beta}_2 x_{i2} + \cdots) = \hat{\beta}_0\underbrace{\sum \hat{\varepsilon}_i}_{0} + \hat{\beta}_1\underbrace{\sum \hat{\varepsilon}_i x_{i1}}_{0} + \hat{\beta}_2\underbrace{\sum \hat{\varepsilon}_i x_{i2}}_{0} + \cdots = 0$$

So the cross terms are 0 and $SST = SS_{reg} + RSS$. □

*Proof of $SST = SS_{reg} + RSS$ (vectors/matrices).* Define $\|\mathbf{a}\|^2 = \sum_{k=1}^{\ell} a_k^2 = \mathbf{a}\cdot\mathbf{a} = \mathbf{a}'\mathbf{a}$ and $\mathbb{1} = (1,\ldots,1)'$. We want

$$\underbrace{\|\mathbf{Y} - \bar{y}\mathbb{1}\|^2}_{SST} = \underbrace{\|\hat{\mathbf{Y}} - \bar{y}\mathbb{1}\|^2}_{SS_{reg}} + \underbrace{\|\hat{\boldsymbol{\epsilon}}\|^2}_{RSS}$$

Expanding,

$$\|(\mathbf{Y} - \hat{\mathbf{Y}}) + (\hat{\mathbf{Y}} - \bar{y}\mathbb{1})\|^2 = \|\mathbf{Y} - \hat{\mathbf{Y}}\|^2 + \|\hat{\mathbf{Y}} - \bar{y}\mathbb{1}\|^2 + \underbrace{2(\mathbf{Y} - \hat{\mathbf{Y}})'(\hat{\mathbf{Y}} - \bar{y}\mathbb{1})}_{\text{show } =0}$$

and with $\mathbf{Y} - \hat{\mathbf{Y}} = \hat{\boldsymbol{\epsilon}}$, $\hat{\mathbf{Y}} = \mathbf{X}\hat{\boldsymbol{\beta}}$:

$$2\hat{\boldsymbol{\epsilon}}'(\hat{\mathbf{Y}} - \bar{y}\mathbb{1}) = 2\underbrace{\hat{\boldsymbol{\epsilon}}'\mathbf{X}}_{\mathbf{0}}\hat{\boldsymbol{\beta}} - 2\bar{y}\underbrace{\hat{\boldsymbol{\epsilon}}'\mathbb{1}}_{\sum\hat{\epsilon}_i=0} = 0$$ □

**Definition - Degrees of Freedom**

Each term has a degree of freedom (like the denominator of a sample variance):

- $SST$: $n-1$ *(df when estimating $\mu$ with $\bar{y}$)*
- $RSS$: $n-p-1$, $p$ = number of predictors *(df when fitting the regression model)*
- $SS_{reg}$: $(p+1)-1 = p$ *(we have $p+1$ $\beta_j$'s)*

DFs decompose too: $n-1 = (n-p-1)+p$.

## ANOVA Test for Overall Significance

*Remark* (Why We Need Another Test).

- The $t$-test on $\beta_j$ ($H_0 : \beta_j = 0$, $t^* = \hat{\beta}_j/SE(\hat{\beta}_j)$) in **SLR** is equivalent to testing whether a linear relationship exists in the model.
- In **MLR** it only says whether $X_j$ is linearly related to $Y$ *in the presence of the others* — not whether a linear relationship exists in the model in general.

**Example 61 House Pricing (Simulated)**
True model $Y = 1 + 3X_1 + 2X_2 + \epsilon$, $X_1$ = #bedrooms, $X_2$ = #bathrooms, with $X_1 \approx X_2$ (very correlated):

```
n = 100
set.seed(1234)
numBedrooms  = sample(c(0,1,2,3,4,5), size = n, replace = TRUE)
numBathrooms = numBedrooms + round(runif(n,-0.51,0.51))  # differ by at most 1
sellPrice    = 1 + 3*numBedrooms + 2*numBathrooms + rnorm(n, 0, 3)
model <- lm(sellPrice ~ numBedrooms + numBathrooms)
```

```
             Estimate Std. Error t value Pr(>|t|)
(Intercept)    1.9156     0.5956   3.216  0.00177 **
numBedrooms    2.3661     2.2732   1.041  0.30053
```

```
numBathrooms    2.4111     2.2772   1.059  0.29232
Residual standard error: 3.182 on 97 degrees of freedom
Multiple R-squared: 0.879, Adjusted R-squared: 0.8765
F-statistic: 352.4 on 2 and 97 DF, p-value: < 2.2e-16
```

Neither #beds nor #baths is significant in the presence of the other, but we **know** there is a linear relationship from the data-generating process. Scatterplots of `sellPrice` vs. either predictor show a clear linear trend, and each SLR model agrees:

| Model | $\hat{\beta}_1$ | SE | $t$ | p-value |
|---|---|---|---|---|
| `sellPrice ~ numBedrooms` | 4.7654 | 0.1798 | 26.509 | $< 2 \times 10^{-16}$ |
| `sellPrice ~ numBathrooms` | 4.7740 | 0.1800 | 26.515 | $< 2 \times 10^{-16}$ |

Unsurprising since $X_1 \approx X_2$. How do we test, in the MLR model, whether at least one predictor explains enough of the variation in $Y$? **Use the sum of squares decomposition.**

*Remark* (Building a Test from the Decomposition).

- Significant linear relationship = a significant amount of the variation in the response was explained by the model: $SS_{reg}$ large, explaining a large portion of $SST$.
- A good model has $SS_{reg}$ much larger than $RSS$; by the decomposition, $SS_{reg}/SST$ large necessarily means $RSS/SST$ small.
- The test of overall significance checks whether $SS_{reg}$ is statistically significantly bigger than $RSS$.
- Every dataset has a different $SST$, so we can't look at $SS_{reg}/SST$ directly.

**Definition - ANOVA Test of Overall Significance**

Split $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1)'$ and test

$$H_0 : \boldsymbol{\beta}_1 = \mathbf{0} \;\; (\beta_1 = \beta_2 = \cdots = \beta_p = 0) \quad \text{vs.} \quad H_A : \boldsymbol{\beta}_1 \neq \mathbf{0} \;\; (\text{at least one slope} \neq 0)$$

Assuming $H_0$ is true,

$$F^* = \frac{SS_{reg}/p}{RSS/(n-p-1)} = \frac{MS_{reg}}{MSR} \sim F(p,\, n-p-1)$$

with $SS_{reg} = \sum(\hat{y}_i - \bar{y})^2$, $RSS = \sum(y_i - \hat{y}_i)^2$. *$F(df_1, df_2)$: numerator df, denominator df.* Dividing each SS by its df standardizes it, giving the **Mean Squares Regression** $MS_{reg}$ and the **Mean Squares Residual** $MSR$.

| Source | DF | Sum of Squares | Mean Squares | $F$ value |
|---|---|---|---|---|
| Regression | $p$ | $SS_{reg}$ | $MS_{reg} = SS_{reg}/p$ | $MS_{reg}/MSR$ |
| Residual | $n-p-1$ | $RSS$ | $MSR = RSS/(n-p-1)$ | – |
| Total | $n-1$ | $SST$ | – | – |

*ANOVA table summarizing the test components.*

*Remark* (Aside: ANOVA for Group Means (STA288)). ANOVA tests group differences: $H_0 : \mu_1 = \mu_2 = \cdots = \mu_k$ (same mean in all $k$ categories) vs. $H_A$: at least one $\mu_j$ differs, using the same $F$ statistic. This is a linear regression! For `lm(y~x)` with one categorical $X$:

$$Y = \beta_0 + \beta_1 I(X = \text{Group } 2) + \cdots + \beta_{k-1} I(X = \text{Group } k) + \epsilon$$

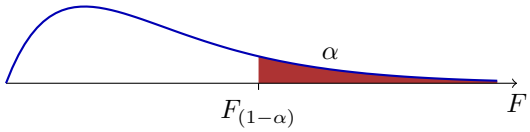
Testing $H_0 : \beta_1 = \cdots = \beta_{k-1} = 0$ with the overall ANOVA test is the same as $H_0 : \mu_1 = \cdots = \mu_k$, since $\beta_0 = \mu_1$ (and $\beta_j = \mu_{j+1} - \mu_1$). *i.e. $Y \sim$ group membership.*

**Definition - $F$ Distribution**

If $U_1 \sim \chi^2_{df_1}$ and $U_2 \sim \chi^2_{df_2}$ (independent), then

$$W = \frac{U_1/df_1}{U_2/df_2} \sim F(df_1, df_2)$$

*Recall: if $Z_i \overset{iid}{\sim} N(0,1)$, then $\sum_{i=1}^k Z_i^2 \sim \chi^2_k$.* $F$ is non-negative, with parameters $df_1, df_2 > 0$; its shape varies with the degrees of freedom.
Under $H_0$ (no linear relationship), $F^* \sim F(p, n-p-1)$ and we expect **small** values: $F^*$ is small if the model doesn't explain much ($SS_{reg}$ small). More extreme values are in the **right tail** (due to $MS_{reg} > MSR$): the larger the ratio, the larger $F^*$ and the smaller the p-value.




*Only the right tail is used.*

**Definition - Concluding the ANOVA Test**

If p-value $< \alpha$, or $F^* > F_{(1-\alpha),(p,n-p-1)}$, reject $H_0$ and conclude a statistically significant linear relationship exists for **at least one** predictor.

**Example 62 House Pricing, Manually**
$H_0 : \beta_1 = \beta_2 = 0$ vs. $H_A$: at least one $\beta_j \neq 0$.

1. **Degrees of freedom**: $n = 100$, $p = 2$; $df_{RSS} = 100 - 2 - 1 = 97$, $df_{SST} = 99$, $df_{SS_{reg}} = 2$.
2. **Sums of squares** (R for the middle steps):

$$SST = (n-1)s_y^2 = 99(81.99404) = 8117.41, \qquad RSS = (n-p-1)s^2 = 982.114,$$

$$SS_{reg} = SST - RSS = 7135.296$$

```
var(sellPrice)                              # 81.99404
sum((model$fitted.values - sellPrice)^2)    # 982.114
```

3. **Mean squares**: $MSR = \frac{982.114}{97} = 10.12489$, $MS_{reg} = \frac{7135.296}{2} = 3567.648$.

4. **Test statistic**: $F^* = \frac{3567.648}{10.12489} = 352.36$.
5. **Conclusion** (p-value or critical value):

```
pf(352.36, 2, 97, lower.tail = FALSE)   # F*, df1, df2, right tail only
# 3.260824e-45
qf(0.95, 2, 97)                         # 3.090187
```

**Reject** $H_0$: a significant linear relationship exists for at least one predictor.

*Also* $R^2 = SS_{reg}/SST = 7135.296/8117.41 = 0.879$*, matching the summary.* In R, the overall test is the last line of `summary()`: `F-statistic: 352.4 on 2 and 97 DF, p-value: < 2.2e-16` ($F^*$, $df_1$, $df_2$).

### Partial $F$ Test

*Remark* (Testing a Subset of Predictors). Suppose a model with $X_1, X_2, X_3$: the overall ANOVA test says at least one predictor is related to $Y$, and the $t$-tests say only $X_3$ is significant. Is the simple model with only $X_3$ as good as the 3-predictor model?

- Can we remove the two insignificant predictors **at the same time**? Since each predictor is conditionally related to $Y$, the $t$-test only says you can remove them one at a time (not together).
- A **partial $F$ test** tells us whether both can be removed simultaneously. If the simple model is as good, $SS_{reg}$ and $RSS$ should be similar between the two models; if the reduced model's RSS is too much bigger, it explains less of the variation in $Y$ and we can't remove all the insignificant predictors at once.

**Definition - Full and Reduced Models**

A partial $F$ test compares 2 models fit to the same data/response (so $SST$ is the same):

- **Complete/full** model (the bigger one) with $p$ predictors;
- **Reduced** model (the smaller one) with $p - k$ predictors ($k$ removed).

A model with more predictors (even non-significant ones) has smaller RSS:

$$RSS_{reduced} \geq RSS_{full} \iff SS_{reg,reduced} \leq SS_{reg,full}$$

so we adjust for the additional predictors by standardizing with each model's df.

**Definition - Partial $F$ Test**

Write $\boldsymbol{\beta} = (\beta_0, \boldsymbol{\beta}_1, \boldsymbol{\beta}_2)'$, where $\boldsymbol{\beta}_1$ holds the $p - k$ coefficients to **keep** and $\boldsymbol{\beta}_2$ the $k$ coefficients we **consider dropping**. Test

$$H_0 : \boldsymbol{\beta}_2 = \mathbf{0} \quad (\beta_{p-k+1} = \cdots = \beta_p = 0) \quad \text{vs.} \quad H_A : \boldsymbol{\beta}_2 \neq \mathbf{0} \quad (\text{at least one} \neq 0)$$

Quantify the difference in RSS:

$$RSS_{drop} = RSS_{reduced} - RSS_{full} = SS_{reg,full} - SS_{reg,reduced}$$

- Full model better: it explains more of the variation in $Y$, $RSS_{full}$ is much smaller than $RSS_{reduced} \Rightarrow$ **bigger** $RSS_{drop}$.
- Similar models: a similar amount of variation is explained, the RSS's are similar $\Rightarrow$ **smaller** $RSS_{drop}$.

Like the ANOVA test, compare the mean SS difference to the model with the smallest mean squares residual (the full model):

$$F^* = \frac{RSS_{drop}/k}{RSS_{full}/(n-p-1)} \sim F(k,\, n-p-1) \text{ under } H_0$$

*If the reduced model is $y \sim 1$, this is exactly the ANOVA overall test statistic.*

### Concluding the Partial $F$ Test

$H_0$ says the larger (full) model is not better than the smaller (reduced) model ($MSR_{drop}$ small relative to $MSR_{full}$).

- If $F^* < F_{(1-\alpha),(k,\,n-p-1)}$: **fail to reject** $H_0$. The additional $SS_{reg}$ from the $k$ predictors isn't enough to keep them *(the outcome we want if aiming for simpler models)*. Conclude no significant linear relationship between $Y$ and <u>any</u> of the $k$ predictors.
- If $F^* > F_{(1-\alpha),(k,\,n-p-1)}$ or p-value $< \alpha$: **reject** $H_0$. The additional $SS_{reg}$ explains a lot of variation, so keep these predictors. Conclude a significant linear relationship exists between $Y$ and **<u>at least one</u>** of the $k$ predictors we tried dropping.

**Example 63 Iris Dataset**
Edgar Anderson's iris data (built into R as `iris`): 150 flowers, 50 from each of 3 species (setosa, versicolor, virginica), with sepal/petal length and width in cm. Model petal length by petal width and species:

$$\text{Length} = \beta_0 + \beta_1 \text{Width} + \beta_2 I(\text{Species} = \text{versicolor}) + \beta_3 I(\text{Species} = \text{virginica}) + \epsilon$$

```
model2 <- lm(Petal.Length ~ Petal.Width + Species, data = iris)
```

```
                  Estimate Std. Error t value Pr(>|t|)
(Intercept)        1.21140    0.06524  18.568  < 2e-16 ***
Petal.Width        1.01871    0.15224   6.691 4.41e-10 ***
Speciesversicolor  1.69779    0.18095   9.383  < 2e-16 ***
Speciesvirginica   2.27669    0.28132   8.093 2.08e-13 ***
Residual standard error: 0.3777 on 146 degrees of freedom
Multiple R-squared: 0.9551, Adjusted R-squared: 0.9542
F-statistic: 1036 on 3 and 146 DF, p-value: < 2.2e-16
```

**Can we omit species?** We can't use the $t$-test since multiple $\beta_j$'s correspond to Species; we want to test $\beta_2 = \beta_3 = 0$ simultaneously ($\beta_1$ for petal width stays). Use the partial $F$ test.

**Example 64 Iris: Partial $F$ Test by Hand**
$H_0 : \beta_2 = \beta_3 = 0$ vs. $H_A$: at least one of $\beta_2, \beta_3 \neq 0$.

1. **Models and RSS**: full `length ~ width + species`, reduced `length ~ width`.

```
y <- iris$Petal.Length
sum((model2$fitted.values - y)^2)           # 20.83344 (full)
sum((model2_reduced$fitted.values - y)^2)   # 33.84475 (reduced)
```

2. $RSS_{drop} = RSS_{reduced} - RSS_{full} = 33.84475 - 20.83344 = 13.01131$. *Always positive ($RSS_{full}$ is always smaller than $RSS_{reduced}$); if negative, check the order of subtraction.* We dropped 2 $\beta$'s, so $k = 2$.
3. **Test statistic**:

$$F^* = \frac{RSS_{drop}/k}{RSS_{full}/(n-p-1)} = \frac{13.01131/2}{20.83344/(150-3-1)} = 45.5914$$

4. **Conclusion**: `pf(45.5914, 2, 146, lower.tail = FALSE)` $= 4.137 \times 10^{-16} \Rightarrow$ **reject** $H_0$: we cannot remove species from the full model.

**Example 65 Iris: Partial $F$ Test Using R**
Fit the model without species as the reduced model, then call `anova(reduced, full)`:

```
model2_reduced <- lm(Petal.Length ~ Petal.Width, data = iris)
anova(model2_reduced, model2)   # anova(reduced, full)

Model 1: Petal.Length ~ Petal.Width
Model 2: Petal.Length ~ Petal.Width + Species
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1    148 33.845
2    146 20.833  2    13.011 45.591 4.137e-16 ***
```

*Even by hand, it helps to organize calculations in a table like this output.*

## Relationships Between the Tests

### Considerations for Using These Tests

- General order (a guideline, not the law): **ANOVA overall test** $\rightarrow$ **$t$-tests on individual coefficients** $\rightarrow$ **partial $F$ test**. The $t$-tests help inform which predictors you might consider dropping.
- Always check assumptions before any hypothesis test; for $F$ tests, verify them in **both** the full and reduced models.
- Cautions for the partial $F$ test:
  - cannot compare models from different datasets;

  - cannot compare models unless one is **nested** in (a subset of) the other — e.g. with model 1: $X_1, X_2, X_3$, model 2: $X_2, X_4$, model 3: $X_2, X_3$, only models 1 and 3 can be compared.
- To remove one categorical predictor with more than 2 categories, we **must** use the partial $F$ test (testing multiple $\beta_j$'s $= 0$ simultaneously).

**Example 66 ANOVA Overall Test as a Partial $F$ Test**
The reduced model is the intercept-only model; same $F^*$ as the `lm()` summary and the manual calculation:

```
anova(lm(sellPrice ~ 1), model)

Model 1: sellPrice ~ 1
Model 2: sellPrice ~ numBedrooms + numBathrooms
  Res.Df    RSS Df Sum of Sq      F    Pr(>F)
1     99 8117.4
2     97  982.1  2    7135.3 352.36 < 2.2e-16 ***
```

*Remark* ($F$-Test vs. $t$-Test).
- $t$-test: $H_0 : \beta_j = 0$ for a single parameter. $F$-test: $H_0 : \beta_1 = \beta_2 = \cdots = 0$, the **joint** significance of multiple predictors.
- Can we run the $t$-test $k$ times instead? **No!** The $F$-test works under multicollinearity (high correlation among predictors); the $t$-test does not.
- For **SLR**, the partial $F$-test, ANOVA overall test and $t$-test of $H_0 : \beta_1 = 0$ are all equivalent.
- Removing only **one** $\beta$ in MLR, the partial $F$-test is equivalent to the $t$-test.

In both equivalent cases, $\boxed{F^* = (t^*)^2}$ with the same p-value, since

$$t = \frac{N(0,1)}{\sqrt{\chi^2_{df}/df}} \quad \Rightarrow \quad t^2 = \frac{\chi^2_1/1}{\chi^2_{df}/df} \sim F(1, df)$$

**Example 67 $F^* = (t^*)^2$ in the House Pricing Data**
**Removing one predictor** from `sellPrice ~ numBedrooms + numBathrooms`:

```
anova(lm(sellPrice ~ numBathrooms), model)
#   Res.Df    RSS Df Sum of Sq      F Pr(>F)
# 1     98 993.08
# 2     97 982.11  1    10.969 1.0834 0.3005
anova(lm(sellPrice ~ numBedrooms), model)
# 1     98 993.46
# 2     97 982.11  1    11.351 1.1211 0.2923
```

matching the $t$-tests: $1.041^2 = 1.0837$ (p $= 0.30053$) for `numBedrooms` and $1.059^2 = 1.1215$ (p $= 0.29232$) for `numBathrooms` *(up to rounding of $t^*$)*.

**SLR**: `anova(lm(sellPrice~1), lm(sellPrice~numBedrooms))` gives $F^* = 702.74$ on 1 and 98 df, the summary's $F$-statistic is 702.7, and $26.509^2 = 702.73$.

*Remark* (Module Takeaways).
- What do the components of the sum of squares decomposition measure, and how are they computed?
- What is the difference between the hypotheses for the ANOVA test and the partial $F$ test?
- How is the decomposition used differently in each, how do we conduct each test, and what are their conclusions?